

Service-oriented Software Engineering (SoSE)

Developing REST WS with Apache
CXF-Maven
(Java 21, Apache CXF 4.1.1)

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Roadmap

- RESTful web services
- Create REST service using Java 21, Apache CXF 4.1.1 and Maven
- Create a java client for invoking our REST service

REST Insight

Focus on SOAP-based web services so far

Now, the question is:

is the full power of SOAP-based Web Services always needed?

Think about the following facts

- service users have not been really interested in dynamic binding features and the use of multiple service providers (as offered, for instance, by UDDI registry)
- current WS standards have been criticized as heavyweight standards that are over-general and inefficient
- service users have been rarely using WS-* standards for, e.g., quality of service, reliability, and so forth

The above facts led to the uptake of RESTful services

REST Insight

RESTful Web Services

REST (REpresentational State Transfer) is an architectural style based on transferring representations of resources from a server to a client

This style is simpler than SOAP/WSDL for implementing web services

RESTful services involve a lower overhead than so-called “big” web services and are used by many organizations implementing “lightweight” service-based systems

REST Insight

Deciding which type of Web Service to use

You want to use **RESTful web services** for integration over the web and **big web services** in enterprise application-integration scenarios with advanced quality-of-service (QoS) requirements.

Jakarta XML Web Services

This is the JAKARTA EE point of view

- Addresses advanced QoS requirements that commonly occur in enterprise computing. When compared to Jakarta RESTful Web Services, XML Web Services makes it easier to support the WS-* set of protocols, which provide standards for security and reliability, among other things, and interoperate with other WS-* conforming clients and servers.

Jakarta RESTful Web Services

- Makes it easier to write web applications that apply some or all of the constraints of the REST style to induce desirable properties in the application, such as loose coupling (evolving the server is easier without breaking existing clients), scalability (start small and grow), and architectural simplicity (use off-the-shelf components, such as proxies or HTTP routers). You would choose to use Jakarta RESTful Web Services for your web application because it is easier for many types of clients to consume RESTful web services while enabling the server side to evolve and scale. Clients can choose to consume some or all aspects of the service and mash it up with other web-based services.

<https://jakarta.ee/learn/docs/jakartaee-tutorial/9.1/websvcs/webservices-intro/webservices-intro.html>

REST Insight

Resources

The fundamental element in a RESTful architecture is a **resource**

Essentially, **a resource is simply a data element** such as a catalog, a medical record, or a document

In general, **resources may have multiple representations**, i.e., they can exist in different formats, e.g., MS WORD, PDF, . . .

Resource **operations**

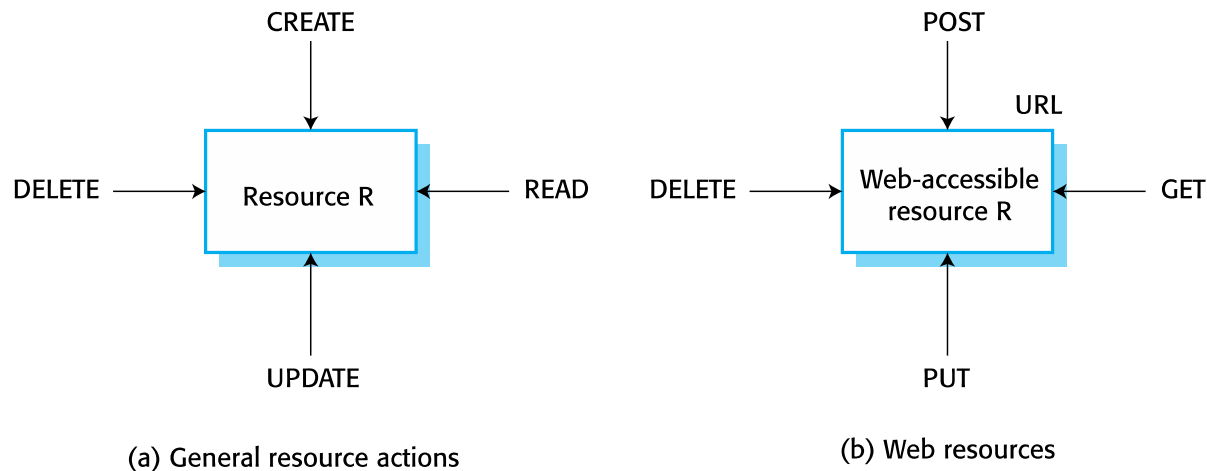
- **create** – bring the resource into existence
- **read** – return a representation of the resource
- **update** – change the value of the resource
- **delete** – make the resource inaccessible

REST Insight

Resources and Actions

The REST style underlies the web as a whole

- **POST** is used to **create** a resource. It has associated data that defines the resource
- **GET** is used to **read** the value of a resource and return that to the requestor in the specified representation, such as XHTML, that can be rendered in a web browser
- **PUT** is used to **update** the value of a resource
- **DELETE** is used to **delete** the resource



REST Insight

Resource Access

When a RESTful approach is used, the data is exposed and is accessed using its URI

HTTP METHODS	Collection resource, such as <code>https://api.example.com/collection</code>	Member resource, such as <code>https://api.example.com/collection/item3</code>
GET	Retrieve the URIs of the member resources of the collection resource in the response body.	Retrieve representation of the member resource in the response body.
POST	Create a member resource in the collection resource using the instructions in the request body. The URI of the created member resource is <i>automatically assigned</i> and returned in the response <i>Location</i> header field.	Create a member resource in the member resource using the instructions in the request body. The URI of the created member resource is <i>automatically assigned</i> and returned in the response <i>Location</i> header field.
PUT	Replace all the representations of the member resources of the collection resource with the representation in the request body, or create the collection resource if it does not exist.	Replace all the representations of the member resource or create the member resource if it does not exist, with the representation in the request body.
DELETE	Delete all the representations of the member resources of the collection resource.	Delete all the representations of the member resource.

https://en.wikipedia.org/wiki/Representational_state_transfer

REST Insight

Pros and Cons

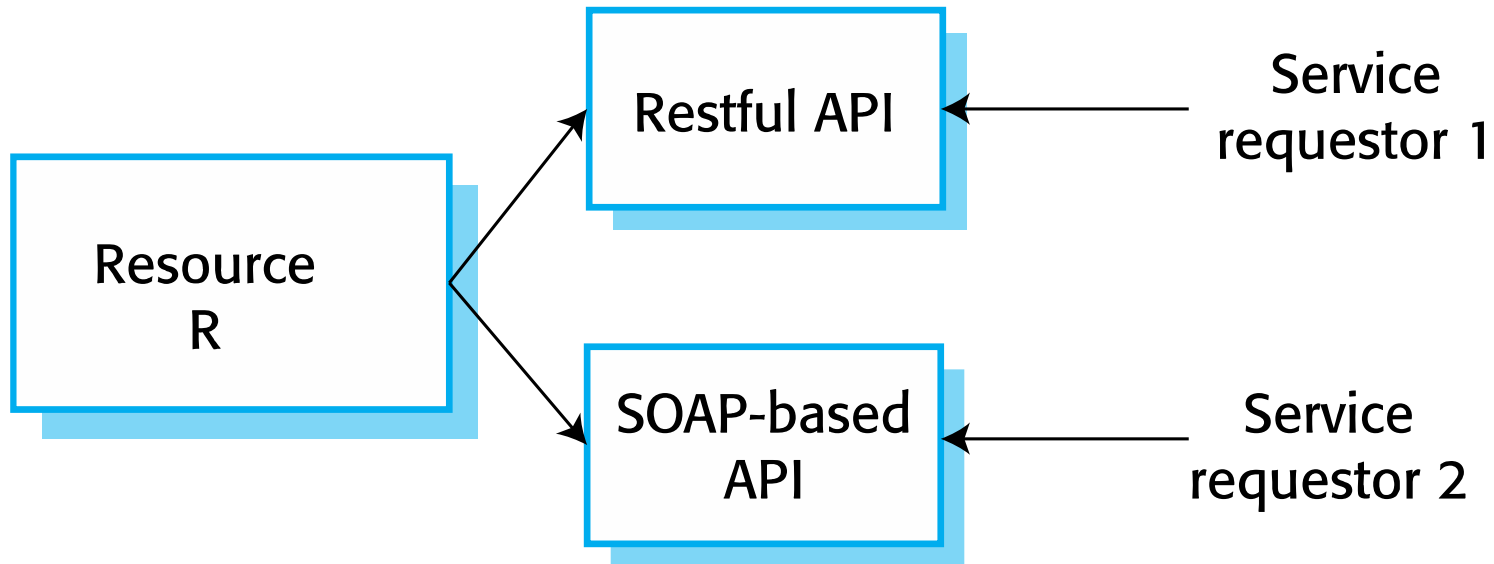
When a service **has a complex interface** and is not a simple resource, it can be (more w.r.t. SOAP Services) **difficult to design a set of RESTful services to represent it**

There are **no standards for RESTful interface description**, so service users must rely on informal documentation to understand the interface

- **WADL** (https://it.wikipedia.org/wiki/Web_Application_Description_Language)
- **OpenAPI** Specification (also known as **Swagger** Specification - https://it.wikipedia.org/wiki/Specifiche_OpenAPI)

REST Insight

RESTful vs SOAP-based APIs



Service Development

Ingredients

Java SE Development Kit 21.0.6

<https://www.oracle.com/java/technologies/javase/jdk21-archive-downloads.html>

Eclipse IDE for Enterprise Java Developers

<https://www.eclipse.org/downloads/packages>

Apache Tomcat® (v.11.0.6) [core - zip or tar.gz]

<https://tomcat.apache.org/download-11.cgi>

Apache CXF™ (v.4.1.1) [binary distribution - zip or tar.gz]

<http://cxf.apache.org/download.html>

Apache Maven (v.3.9.9)

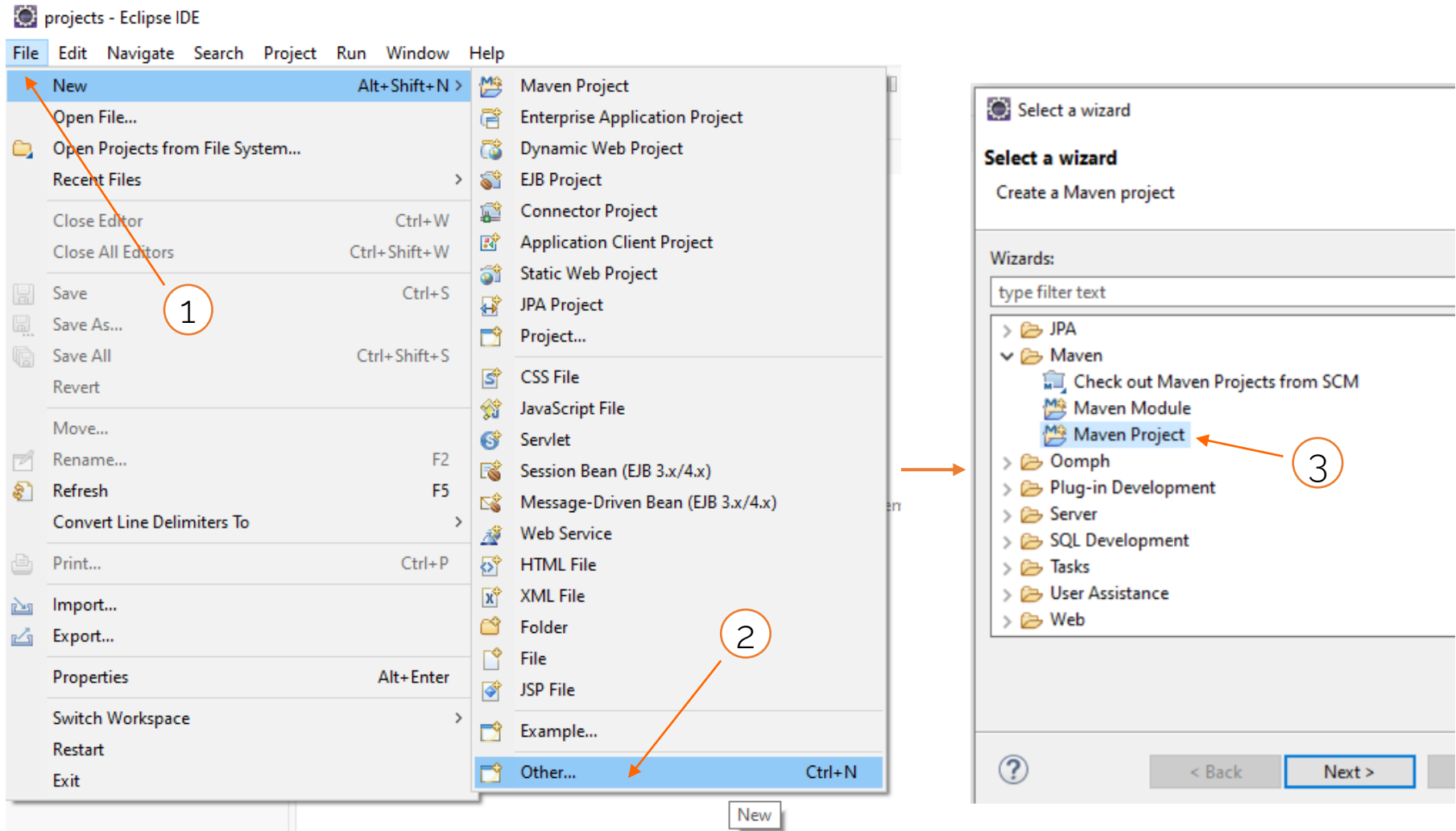
<https://maven.apache.org/download.cgi>

Jakarta Web Services (v4.0.0)

<https://jakarta.ee/specifications/restful-ws/4.0/>

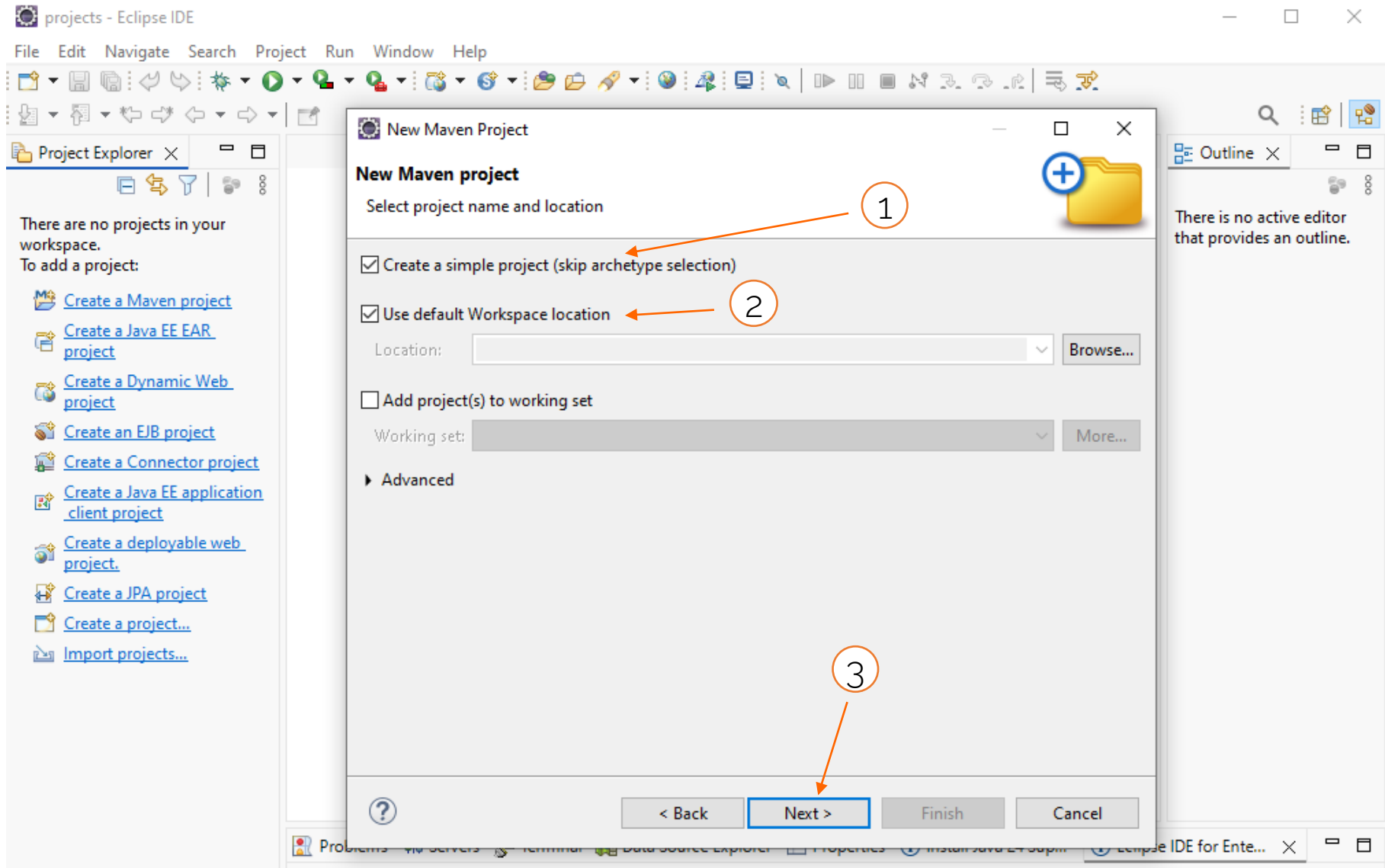
Service Development

New Maven Project (1)



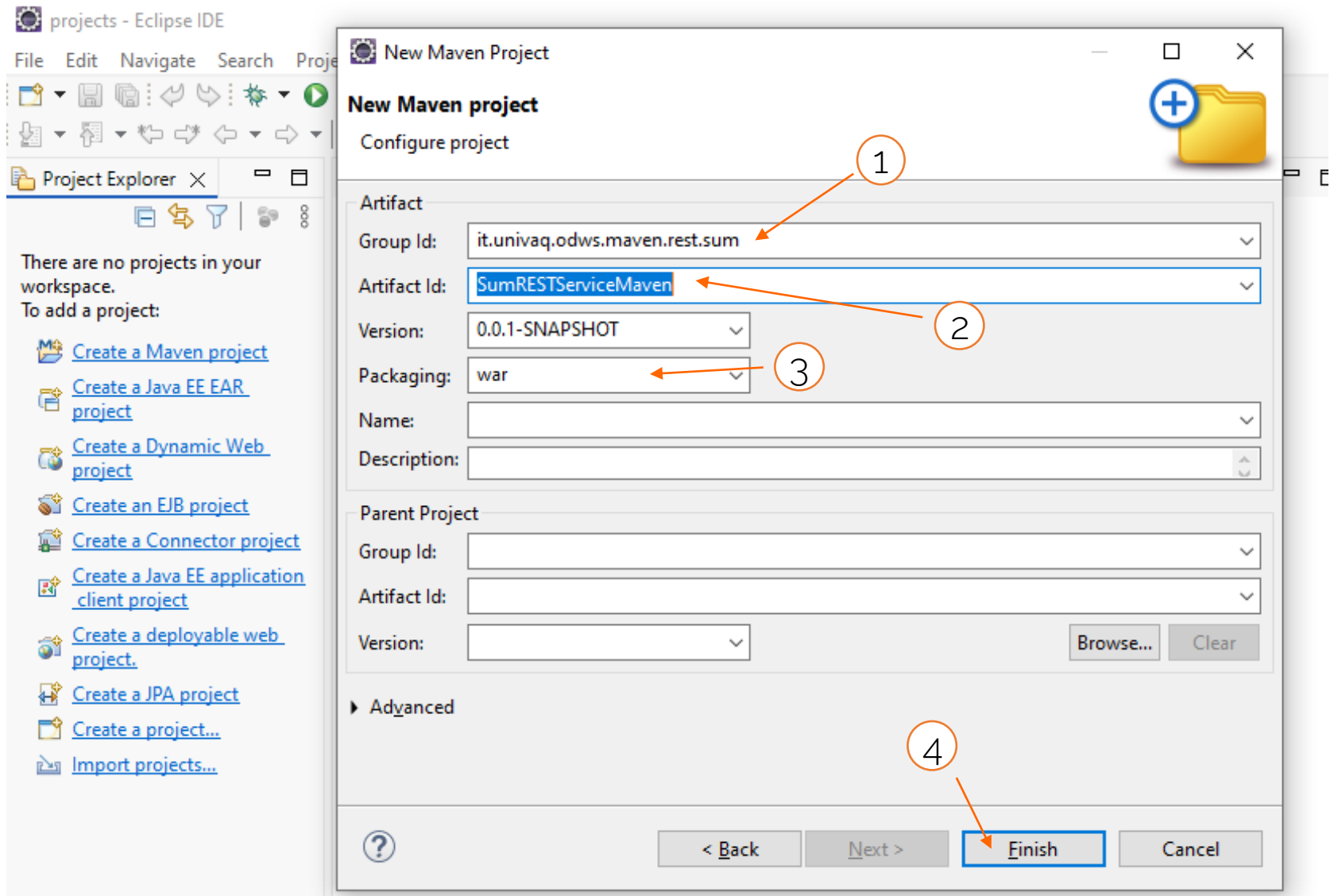
Service Development

New Maven Project (2)



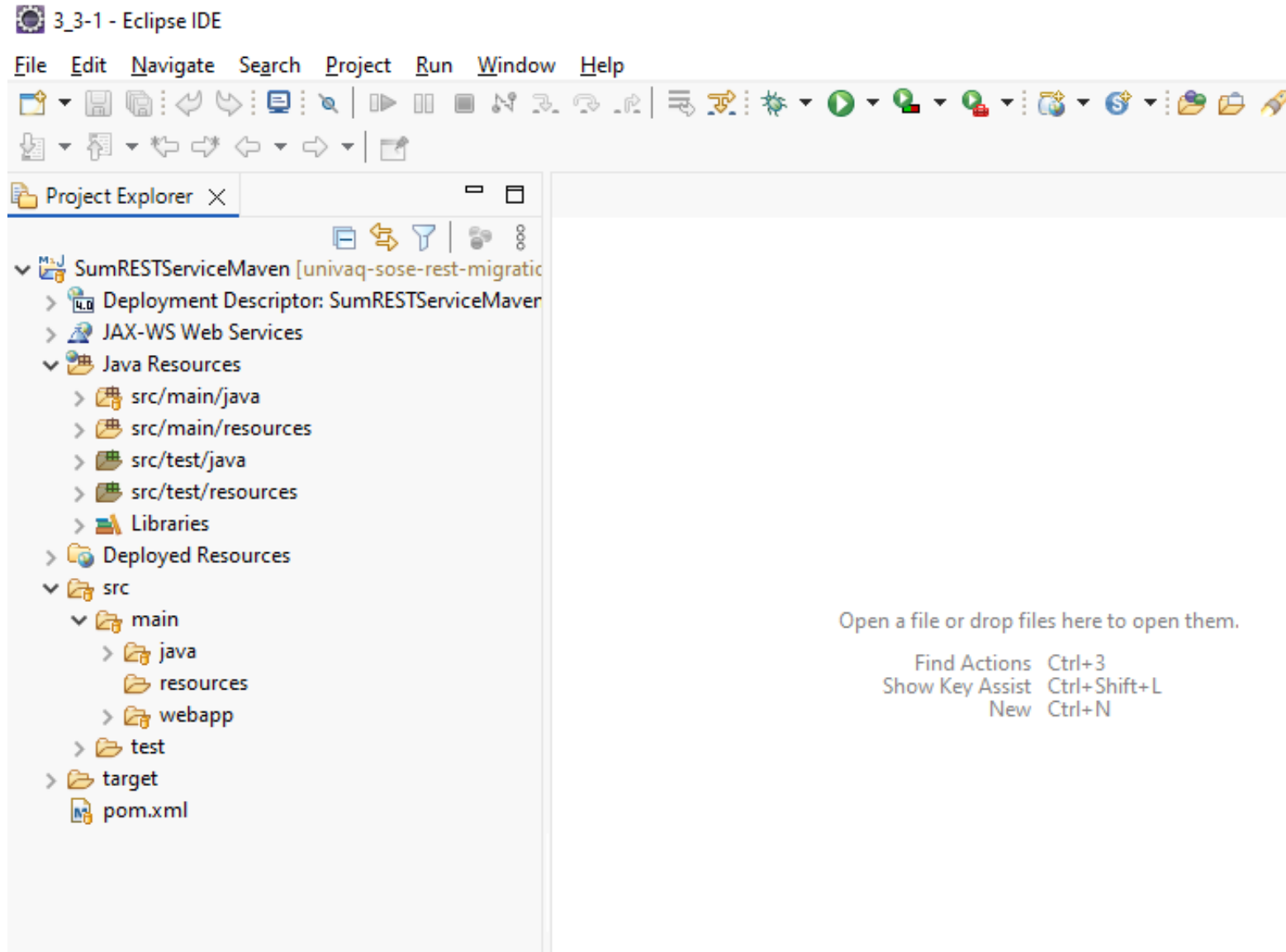
Service Development

New Maven Project (3)



Service Development

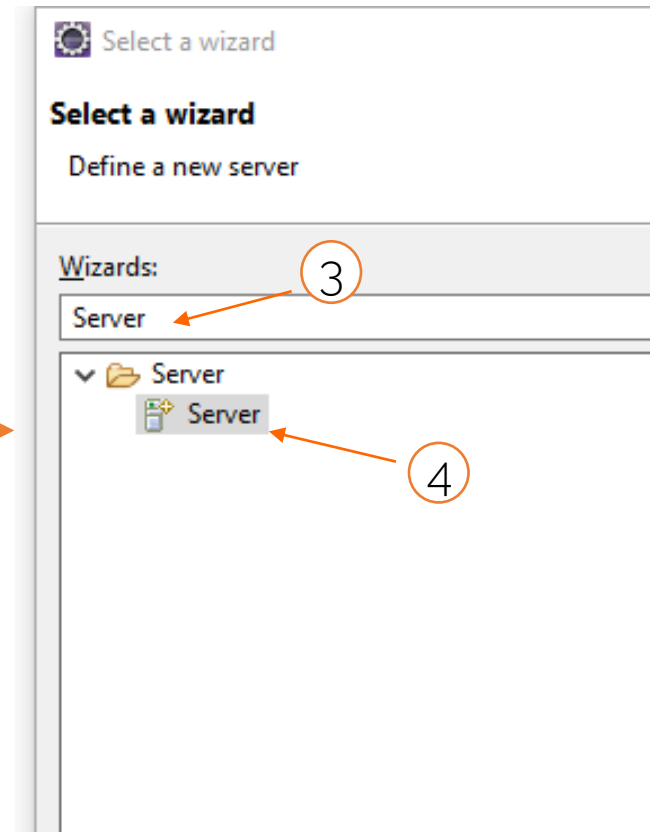
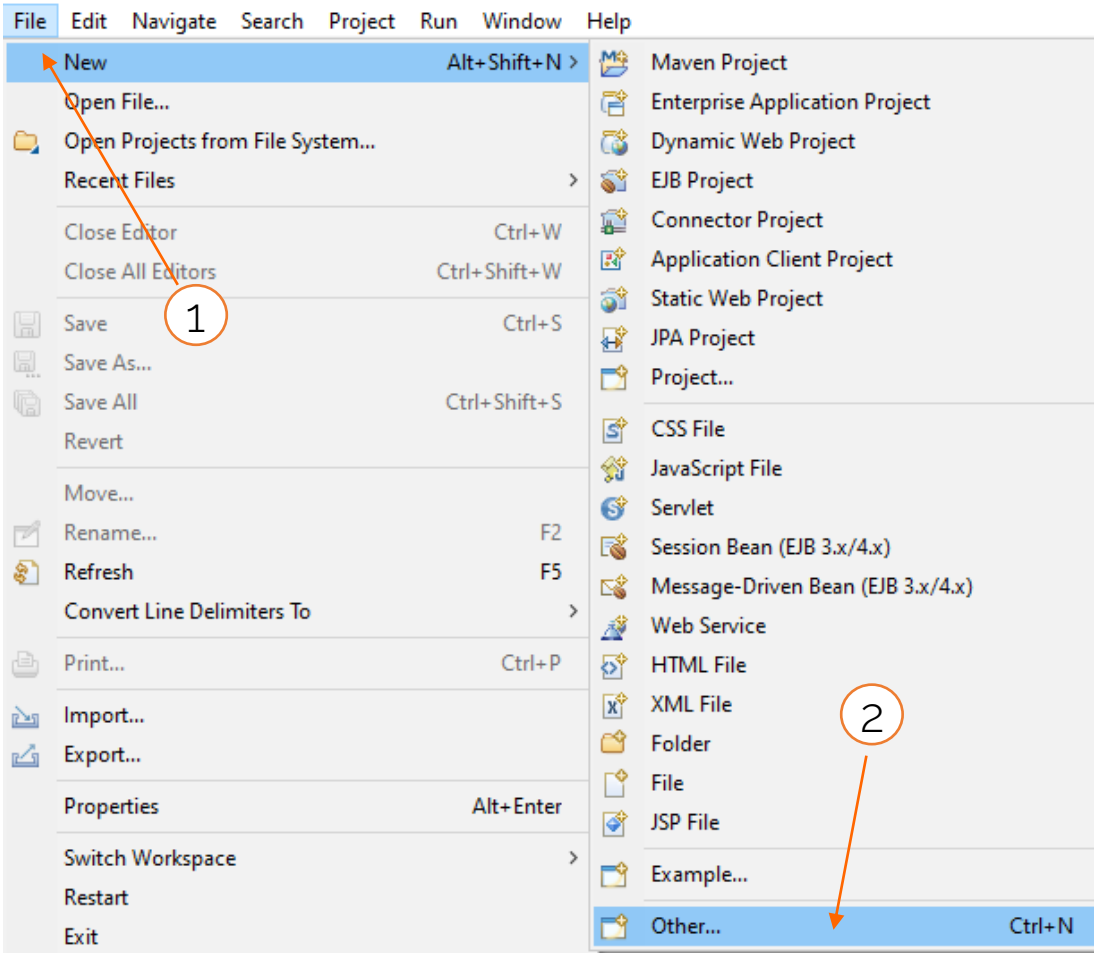
Structure of the Maven Project



Service Development

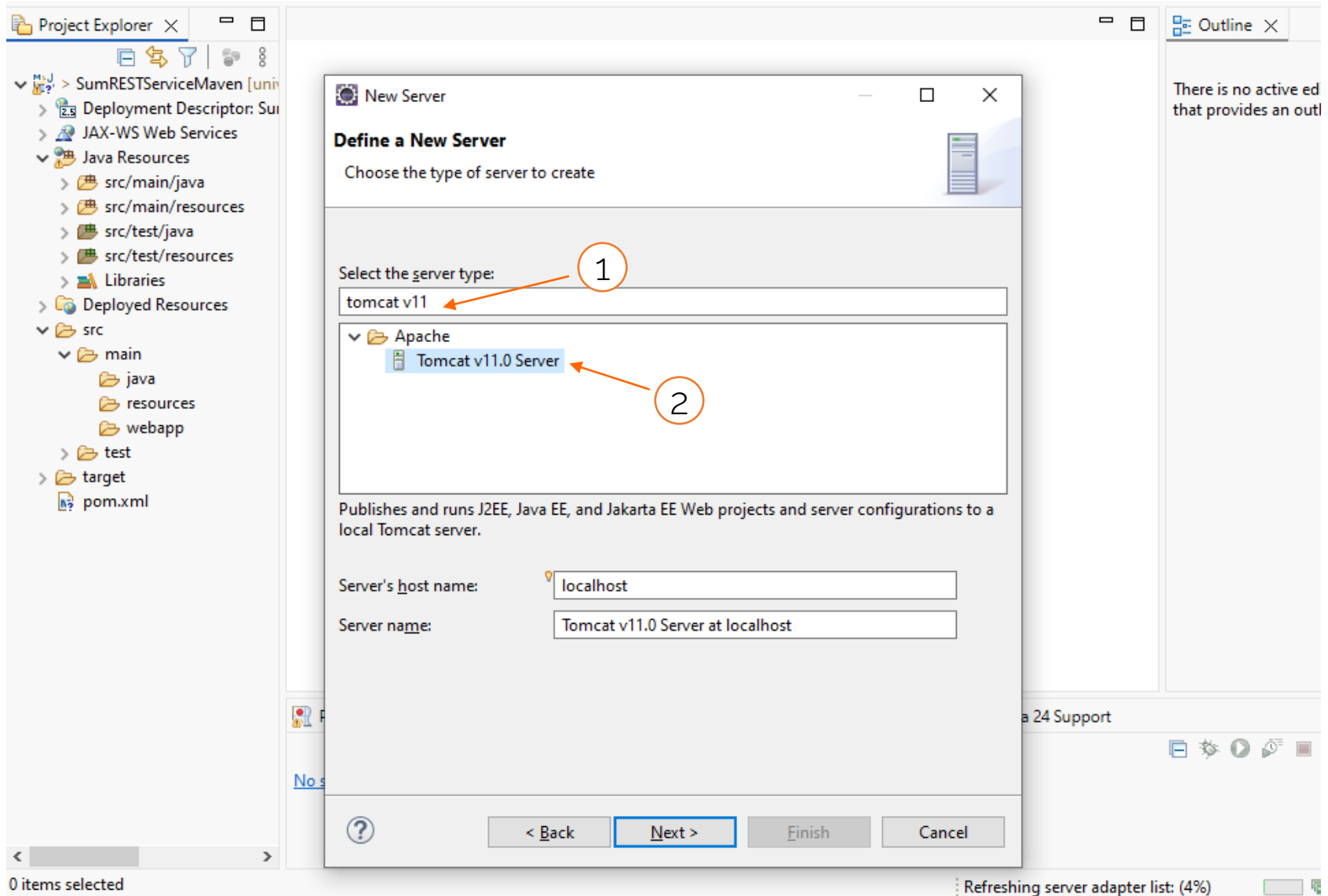
New Server Definition

3_3-1 - Eclipse IDE



Service Development

New Server Definition (2)



Service Development

New Server Definition (3)

New Server

Tomcat Server

Specify the Tomcat installation directory and JRE for this runtime. The specified JRE controls the highest supported Java Facet version.

Name:
Apache Tomcat v11.0

Tomcat installation directory:
D:\Program Files (x64)\apache-tomcat-11.0.6 **Browse...**

apache-tomcat-11.0.4 **Download and Install...**

JRE:
Workbench default JRE
Workbench default JRE
jre
jdk-17.0.3.1
jre1.8.0_431
jdk-17
jre1.8.0_431(2) **Installed JREs...**

1 Select the directory where you installed Tomcat

2

3 Choose the Java 21's jdk if present or select it from Installed JREs... (next slides)

Finish **Cancel**

Service Development

New Server Definition (4)

New Server

Tomcat Server

Specify the Tomcat installation directory and JRE for this runtime. The specified JRE controls the highest supported Java Facet version.

Name:
Apache Tomcat v11.0

Tomcat installation directory:
D:\Program Files (x64)\apache-tomcat-11.0.6 Browse...

apache-tomcat-11.0.4 Download and Install...

JRE:
Workbench default JRE Installed JREs...

- jre
- jdk-17.0.3.1
- jre1.8.0_431
- jdk-17
- jre1.8.0_431(2)

1

? < Back Next > Finish Cancel

Installed JREs

Add, remove or edit JRE definitions. By default, the checked JRE is added to the build path of newly created Java projects.

☒ Detect available JVM installations at startup 2

Installed JREs:

Name	Location
☐ jdk-17	C:\Program Files (x86)\Java\jdk-17
☐ jdk-17.0.3.1	D:\Program Files (x64)\Java Versions\jdk-17.0.3
☒ jre (default)	D:\Program Files (x64)\eclipse enterprise\jdk-17.0.3
☐ jre1.8.0_431	C:\Program Files\Java\jre1.8.0_431
☐ jre1.8.0_431(2)	C:\Program Files (x86)\Java\jre1.8.0_431

2

Add...
Edit...
Duplicate...
Remove
Search...

JRE Type

Select the type of JRE to add to the workspace.

Installed JRE Types:

Execution Environment Description

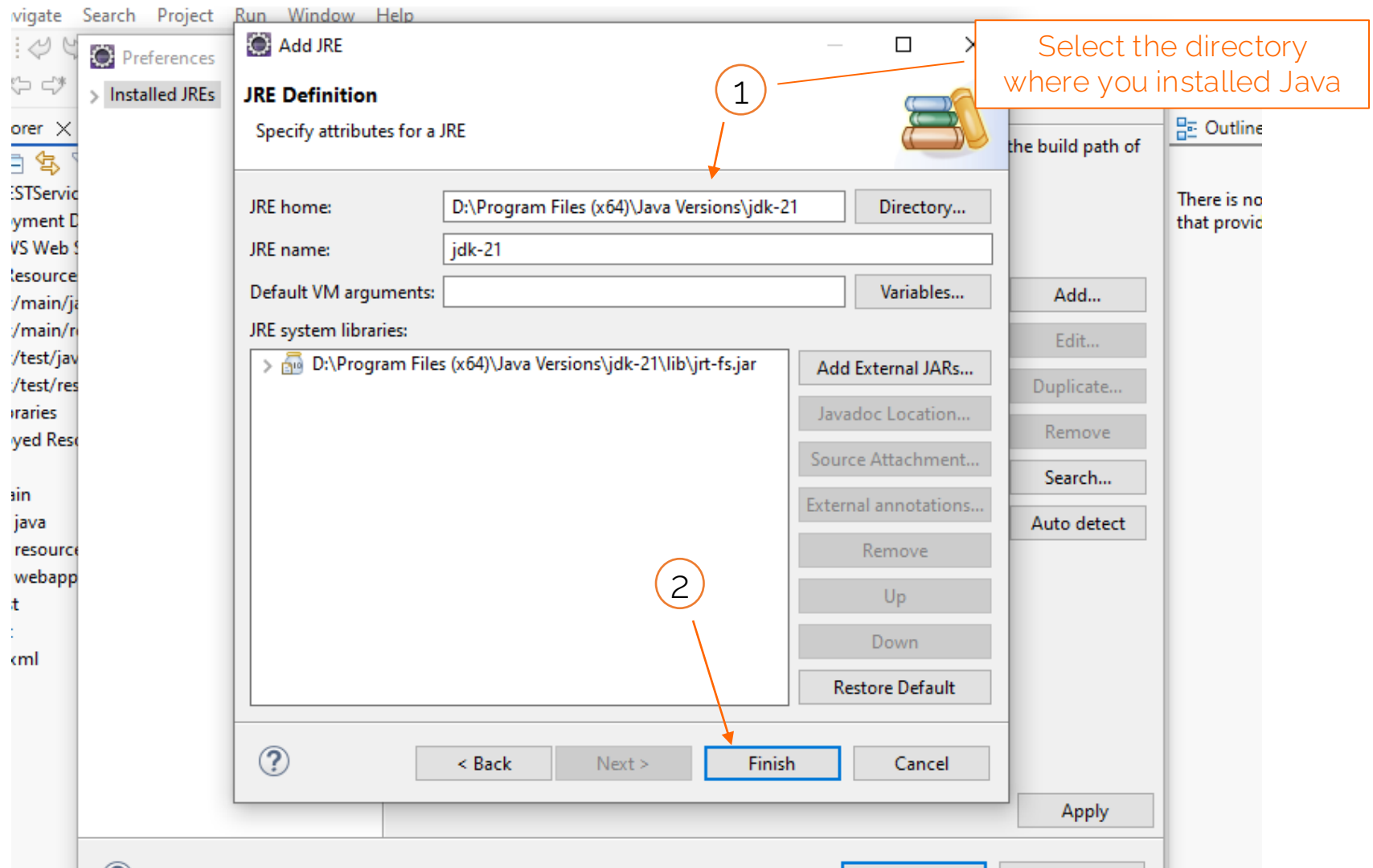
- Standard 1.1.x VM
- Standard VM

3

? < Back Next > Finish Cancel

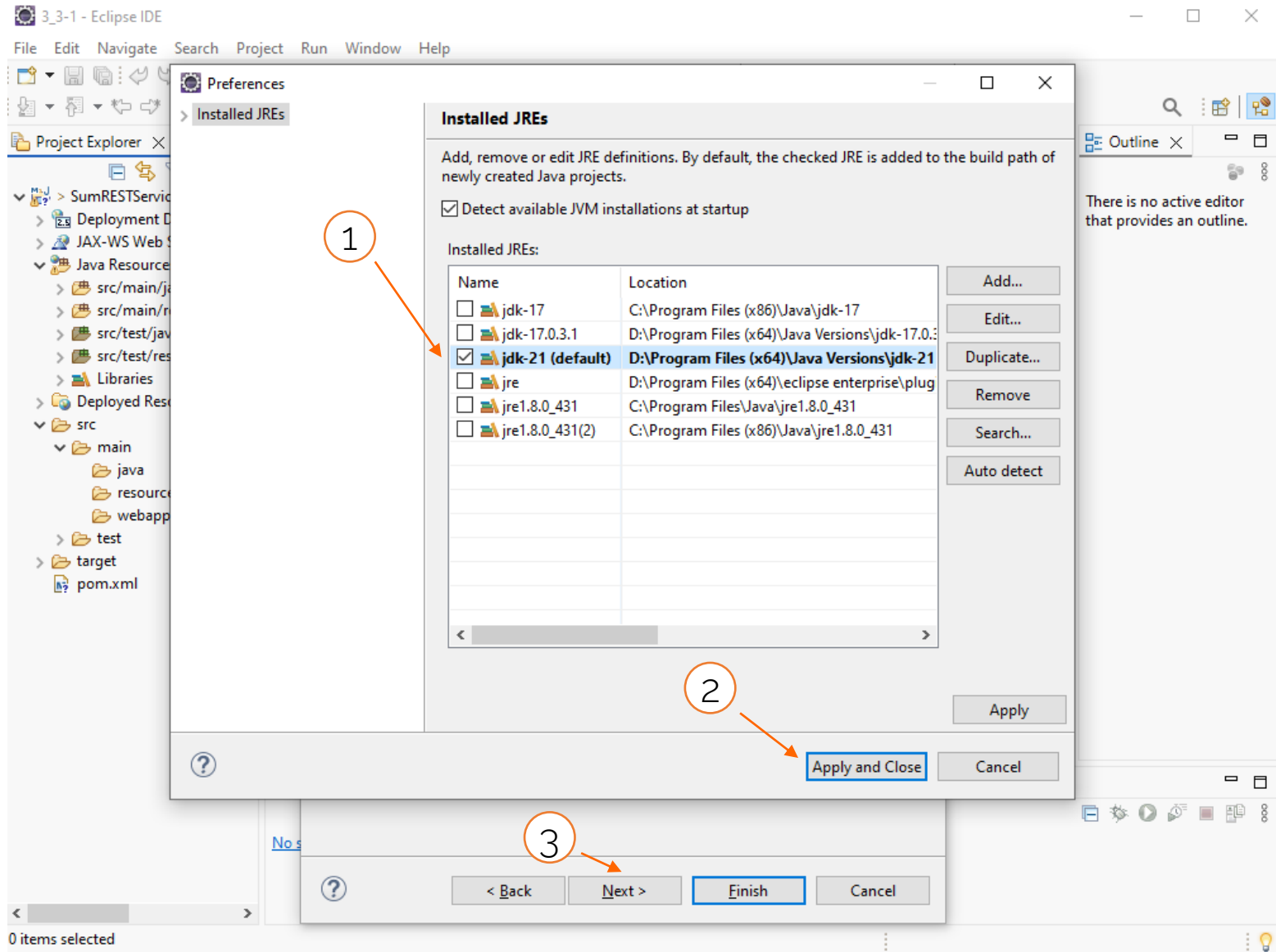
Service Development

New Server Definition (5)



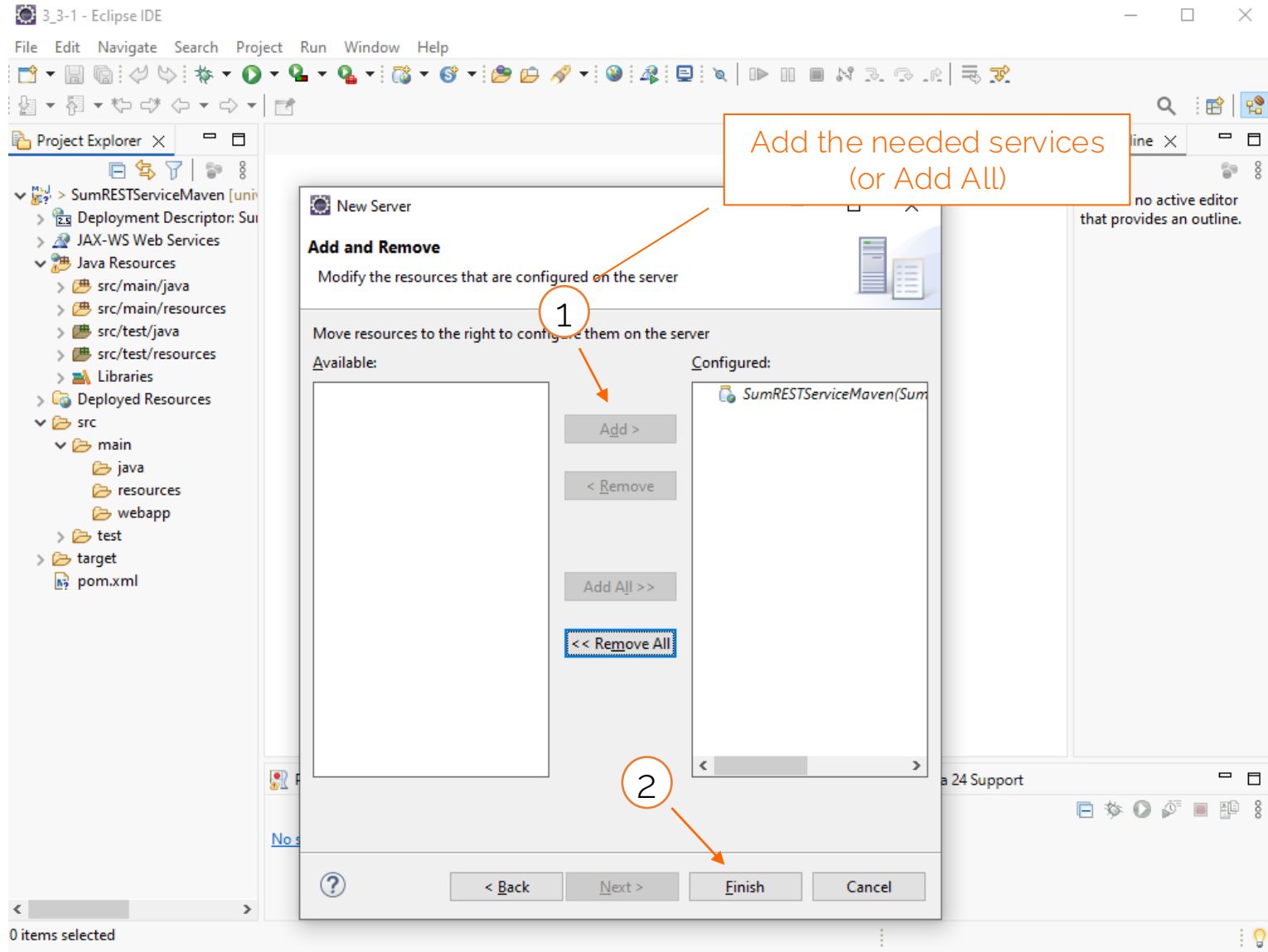
Service Development

New Server Definition (6)



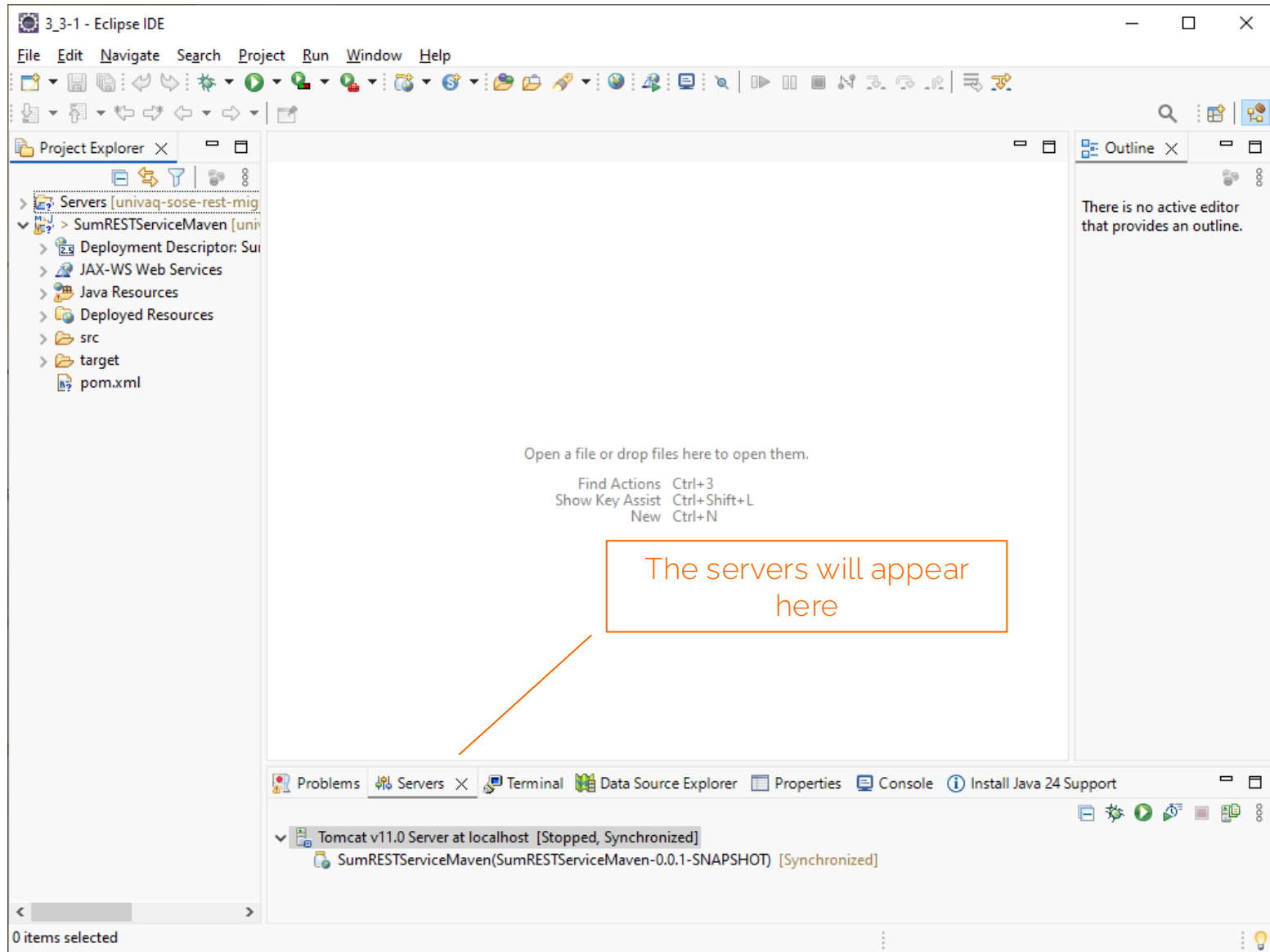
Service Development

New Server Definition (7)



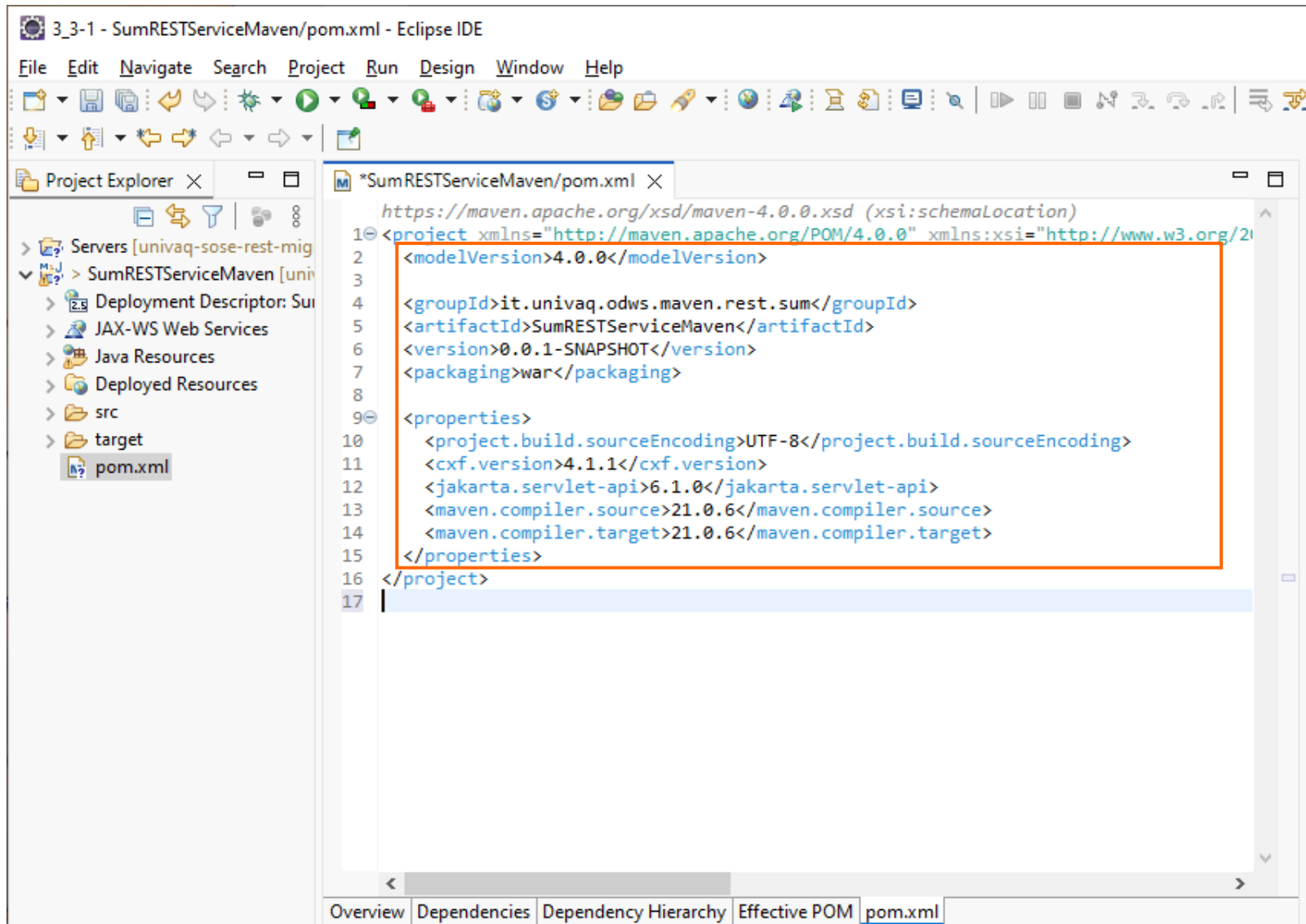
Service Development

New Server Definition (8)



Service Development

Pom Configuration (1)



3_3-1 - SumRESTServiceMaven/pom.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer

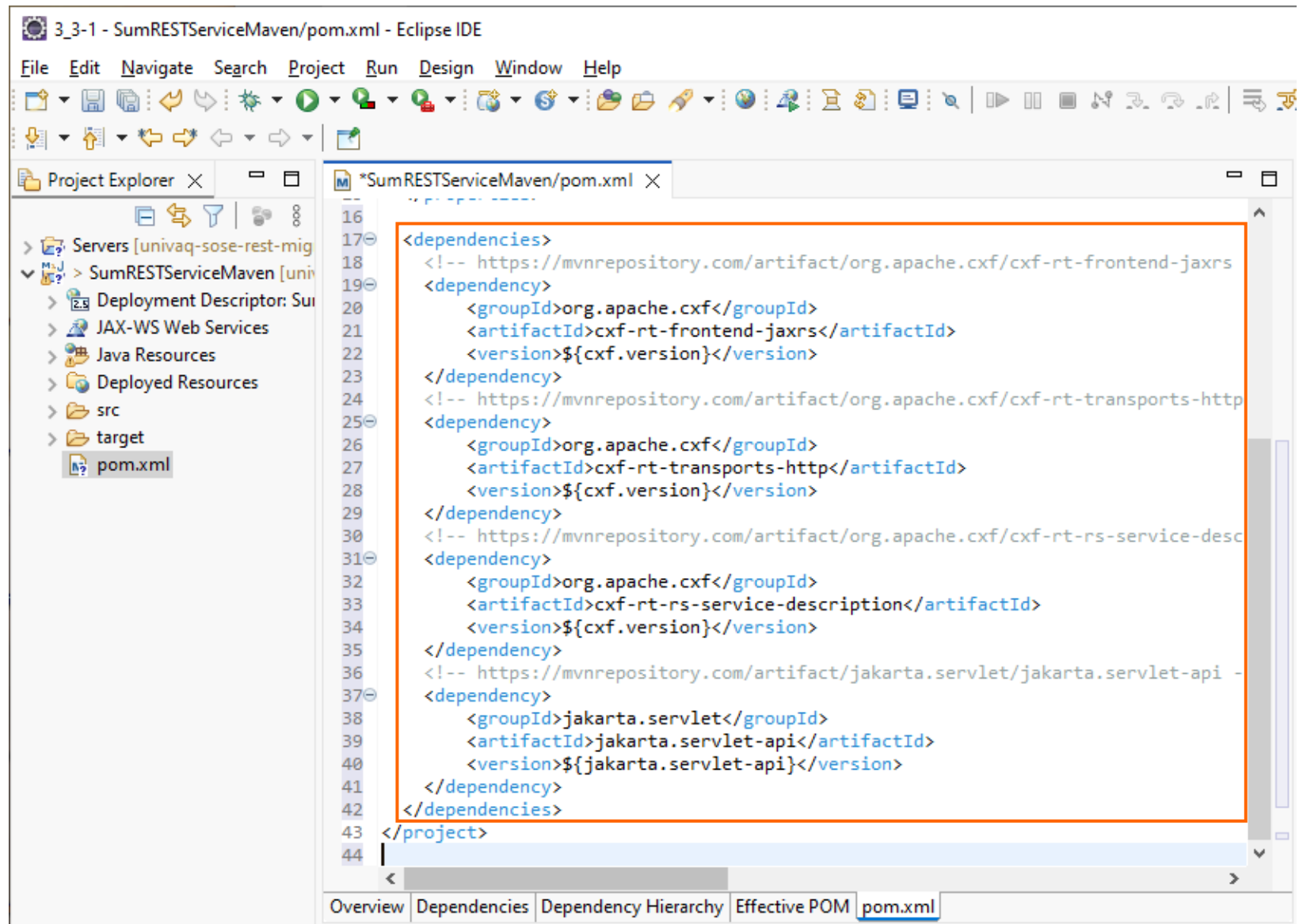
- Servers [univaq-ose-rest-mig]
- SumRESTServiceMaven [univaq-ose-rest-mig]
- Deployment Descriptor: SumRESTServiceMaven
- JAX-WS Web Services
- Java Resources
- Deployed Resources
- src
- target
- pom.xml

```
1 <?xml version="1.0" encoding="UTF-8" ?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3         xsi:schemaLocation="http://maven.apache.org/xsd/maven-4.0.0.xsd http://maven.apache.org/POM/4.0.0.xsd#>
4     <modelVersion>4.0.0</modelVersion>
5     <groupId>it.univaq.odws.maven.rest.sum</groupId>
6     <artifactId>SumRESTServiceMaven</artifactId>
7     <version>0.0.1-SNAPSHOT</version>
8     <packaging>war</packaging>
9     <properties>
10        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
11        <cxfr.version>4.1.1</cxfr.version>
12        <jakarta.servlet-api>6.1.0</jakarta.servlet-api>
13        <maven.compiler.source>21.0.6</maven.compiler.source>
14        <maven.compiler.target>21.0.6</maven.compiler.target>
15    </properties>
16 </project>
17
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

Service Development

Pom Configuration (2)



3_3-1 - SumRESTServiceMaven/pom.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer

- Servers [univaq-sose-rest-mig]
- SumRESTServiceMaven [univaq-sose-rest-mig]
- Deployment Descriptor: SumRESTServiceMaven
- JAX-WS Web Services
- Java Resources
- Deployed Resources
- src
- target
- pom.xml

```
16
17
18 <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-frontend-jaxrs
19 <dependency>
20     <groupId>org.apache.cxf</groupId>
21     <artifactId>cxf-rt-frontend-jaxrs</artifactId>
22     <version>${cxf.version}</version>
23 </dependency>
24 <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-transports-http
25 <dependency>
26     <groupId>org.apache.cxf</groupId>
27     <artifactId>cxf-rt-transports-http</artifactId>
28     <version>${cxf.version}</version>
29 </dependency>
30 <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-rs-service-description
31 <dependency>
32     <groupId>org.apache.cxf</groupId>
33     <artifactId>cxf-rt-rs-service-description</artifactId>
34     <version>${cxf.version}</version>
35 </dependency>
36 <!-- https://mvnrepository.com/artifact/jakarta.servlet/jakarta.servlet-api
37 <dependency>
38     <groupId>jakarta.servlet</groupId>
39     <artifactId>jakarta.servlet-api</artifactId>
40     <version>${jakarta.servlet-api}</version>
41 </dependency>
42 </dependencies>
43 </project>
44
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

Service Development

Pom Configuration (3)

3_3-1 - SumRESTServiceMaven/pom.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer

- Servers [univaq-sose-rest-mig]
- SumRESTServiceMaven [univaq-sose-rest-mig]
- Deployment Descriptor: SumRESTServiceMaven
- JAX-WS Web Services
- Java Resources
- Deployed Resources
- src
- target
- pom.xml

```
32 <groupId>org.apache.cxf</groupId>
33 <artifactId>cxf-rt-rs-service-description</artifactId>
34 <version>${cxf.version}</version>
35 </dependency>
36 <!-- https://mvnrepository.com/artifact/jakarta.servlet/jakarta.servlet-api -->
37 <dependency>
38 <groupId>jakarta.servlet</groupId>
39 <artifactId>jakarta.servlet-api</artifactId>
40 <version>${jakarta.servlet-api}</version>
41 </dependency>
42 </dependencies>
43
44 <build>
45 <plugins>
46 <plugin>
47 <groupId>org.apache.maven.plugins</groupId>
48 <artifactId>maven-compiler-plugin</artifactId>
49 <version>3.14.0</version>
50 <configuration>
51 <source>${maven.compiler.source}</source>
52 <target>${maven.compiler.target}</target>
53 <encoding>${project.build.sourceEncoding}</encoding>
54 </configuration>
55 </plugin>
56 </plugins>
57
58 <finalName>SumRESTServiceMaven</finalName>
59 </build>
60 </project>
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

Service Development

Pom web.xml Possible Error

The screenshot shows the Eclipse IDE interface. The Project Explorer on the left displays the project structure for 'SumRESTServiceMaven', including 'src/main/java', 'src/main/resources', 'src/test/java', 'src/test/resources', 'Libraries', 'JavaScript Resources', 'Deployed Resources', and 'src' (containing 'main', 'test', and 'target' folders). The main editor shows the 'pom.xml' file with the following XML content:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>it.univaq.odws.maven.rest.sum</groupId>
  <artifactId>SumRESTServiceMaven</artifactId>
  <version>0.0.1-SNAPSHOT</version>

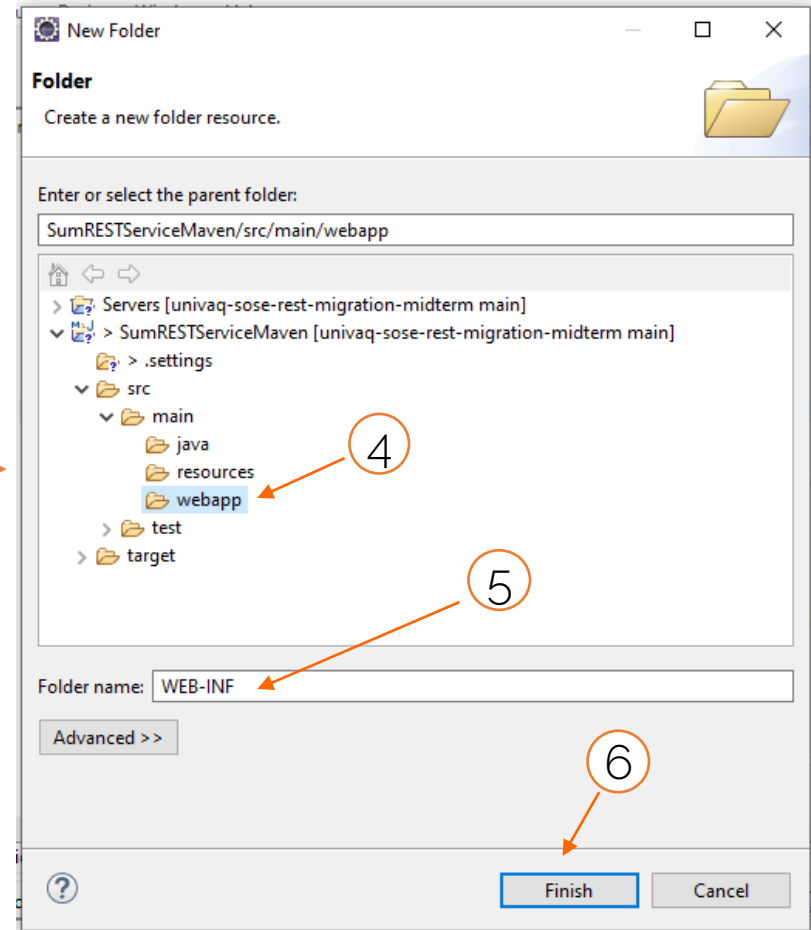
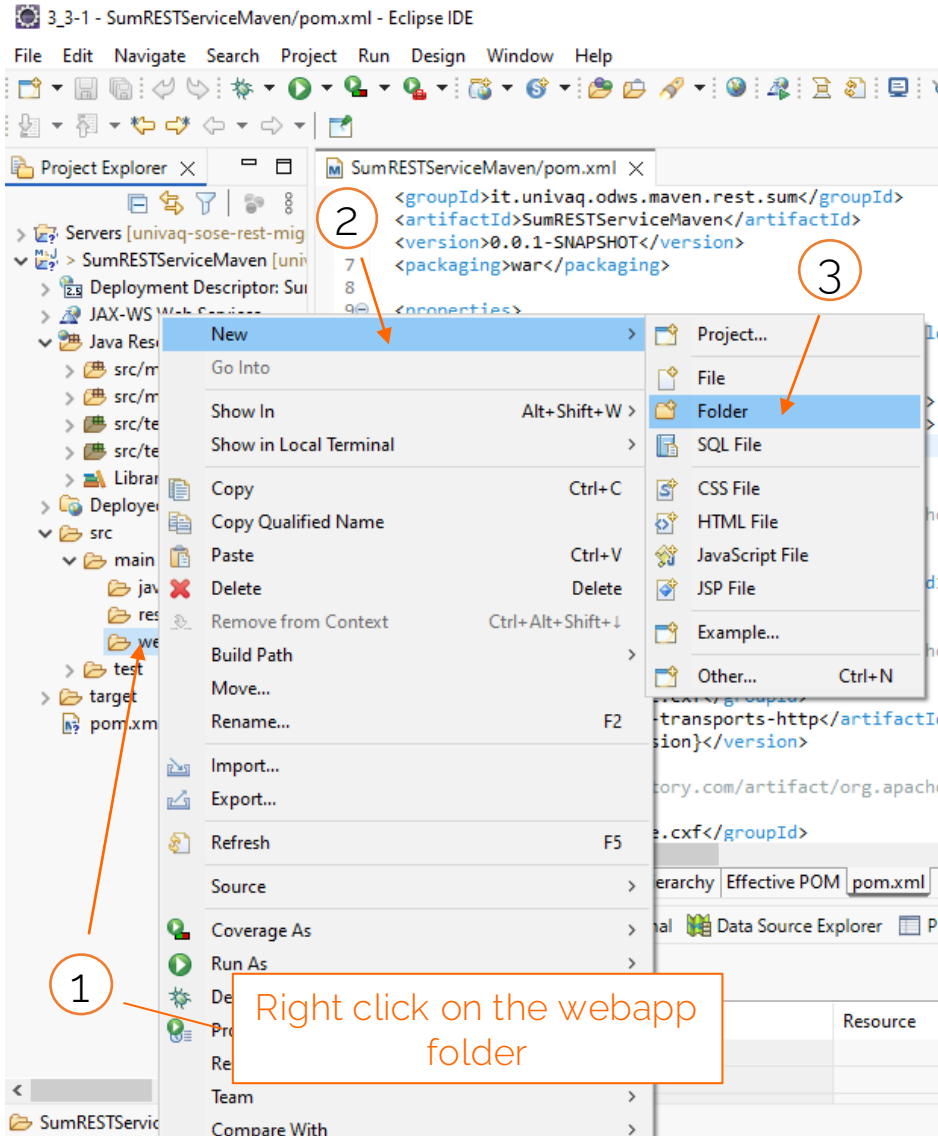
  <webResources>
    <webResource>
      <url>webapp</url>
    </webResource>
  </webResources>

  <properties>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <cxf.version>3.0.0</cxf.version>
  </properties>
</project>
```

A warning icon (red 'x') is present next to line 9, with a tooltip that reads: "web.xml is missing and <failOnMissingWebXml> is set to true". An orange arrow points from this tooltip to a text box that says "It will be fixed in the next slides". The bottom of the IDE shows the 'Servers' tab with 'Tomcat v8.5 Server at localhost [Stopped]'.

Service Development

WEB-INF Folder



Service Development

Copy the web.xml from SumSOAPServiceMaven

3_3-1 - SumRESTServiceMaven/src/main/webapp/WEB-INF/web.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer

- Servers [univaq-rose-rest-mig]
- SumRESTServiceMaven [univaq-rose-rest-mig]
- Deployment Descriptor: SumRESTServiceMaven
- JAX-WS Web Services
- Java Resources
 - src/main/java
 - src/main/resources
 - src/test/java
 - src/test/resources
 - Libraries
- Deployed Resources
- src
 - main
 - java
 - resources
 - webapp
 - WEB-INF
 - web.xml
 - test
 - target
 - pom.xml

web.xml

```
1 http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd (xsi:schemaLocation with catali
2 <?xml version="1.0" encoding="UTF-8"?>
3 <web-app xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4   xmlns="http://xmlns.jcp.org/xml/ns/javaee"
5   xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org/x
6   id="WebApp_ID" version="4.0">
7   <display-name>SumSOAPServiceMaven</display-name>
8
9   <servlet>
10     <display-name>SimpleCXFNonSpringServlet</display-name>
11     <servlet-name>SimpleCXFNonSpringServlet</servlet-name>
12     <servlet-class>it.univaq.odws.maven.sum.SimpleCXFNonSpringServlet</servle
13   </servlet>
14
15   <servlet-mapping>
16     <servlet-name>SimpleCXFNonSpringServlet</servlet-name>
17     <url-pattern>/*</url-pattern>
18   </servlet-mapping>
19
20   <welcome-file-list>
21     <welcome-file>index.html</welcome-file>
22     <welcome-file>index.htm</welcome-file>
23     <welcome-file>index.jsp</welcome-file>
24     <welcome-file>default.html</welcome-file>
25     <welcome-file>default.htm</welcome-file>
26     <welcome-file>default.jsp</welcome-file>
27   </welcome-file-list>
28
```

Design Source

Service Development

Adapt the web.xml to this service

3_3-1 - SumRESTServiceMaven/src/main/webapp/WEB-INF/web.xml - Eclipse IDE

File Edit Source Navigate Search Project Run Window Help

Project Explorer

- Servers [univaq-sose-rest-mig]
- SumRESTServiceMaven [univaq-sose-rest-mig]
- Deployment Descriptor: SumRESTServiceMaven
- JAX-WS Web Services
- Java Resources
 - src/main/java
 - src/main/resources
 - src/test/java
 - src/test/resources
 - Libraries
- Deployed Resources
- src
 - main
 - java
 - resources
 - webapp
 - WEB-INF
 - web.xml
 - test
 - target
 - pom.xml

SumRESTServiceMaven/pom.xml

*web.xml

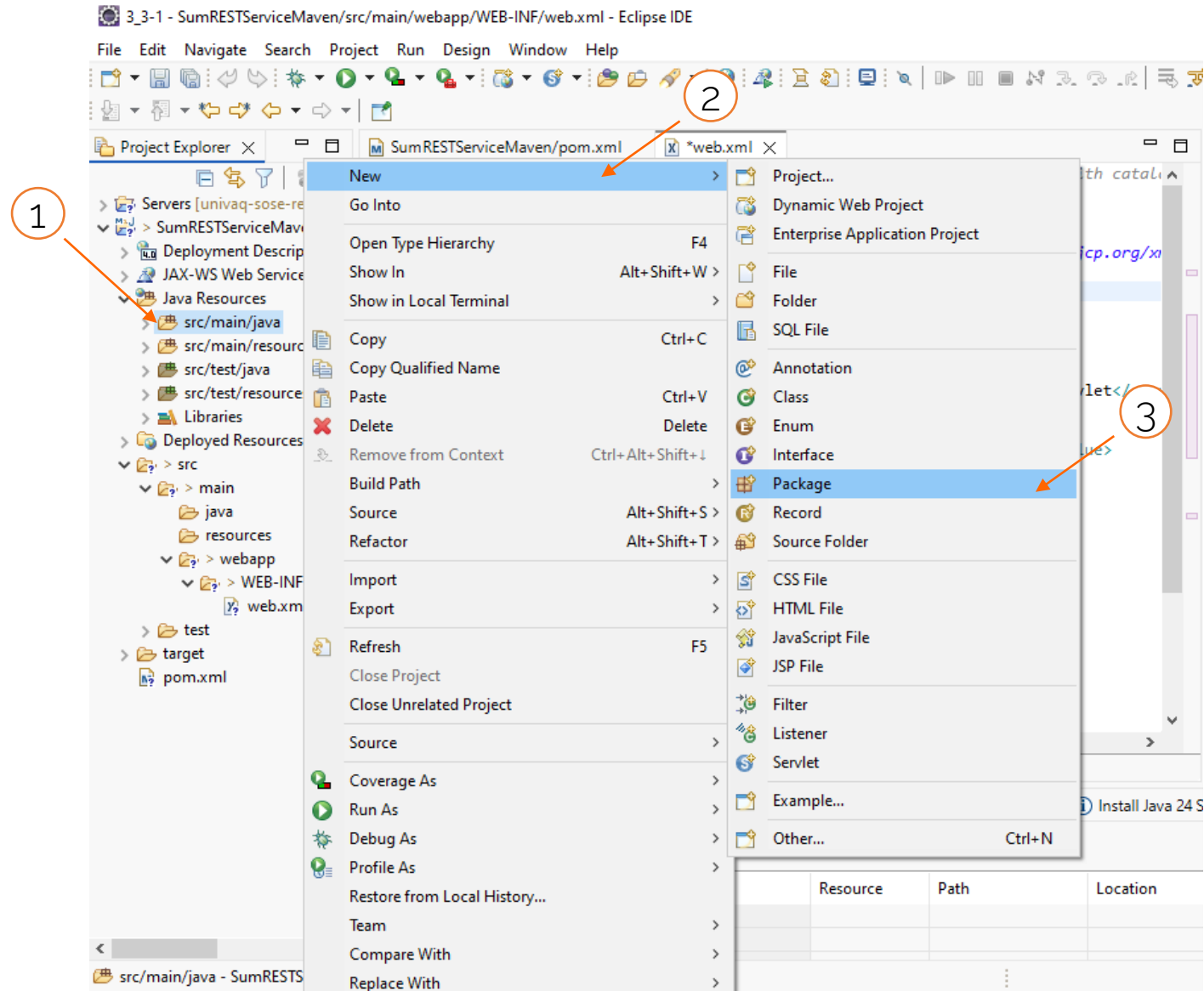
```
http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd (xsi:schemaLocation with catali
1 <?xml version="1.0" encoding="UTF-8"?>
2 <web-app xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xmlns="http://xmlns.jcp.org/xml/ns/javaee"
4   xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org/x
5   id="WebApp_ID" version="4.0">
6
7   <display-name>SumRESTServiceMaven</display-name>
8
9   <servlet>
10     <servlet-name>CXFNonSpringJaxrsServlet</servlet-name>
11     <servlet-class>org.apache.cxf.jaxrs.servlet.CXFNonSpringJaxrsServlet</ser
12     <init-param>
13       <param-name>jaxrs.serviceClasses</param-name>
14       <param-value>it.univaq.odws.maven.rest.sum.SumImpl</param-value>
15     </init-param>
16     <init-param>
17       <param-name>jaxrs.address</param-name>
18       <param-value>/rest</param-value>
19     </init-param>
20   </servlet>
21
22   <servlet-mapping>
23     <servlet-name>CXFNonSpringJaxrsServlet</servlet-name>
24     <url-pattern>*/</url-pattern>
25   </servlet-mapping>
26
27   <welcome-file-list>
28     <welcome-file>index.html</welcome-file>
```

Annotations:

- 1: `xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"`
- 2: `xmlns="http://xmlns.jcp.org/xml/ns/javaee"`
- 3: `xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org/x`
- 4: `id="WebApp_ID" version="4.0">`
- 5: `<init-param>`
- 6: `<init-param>`

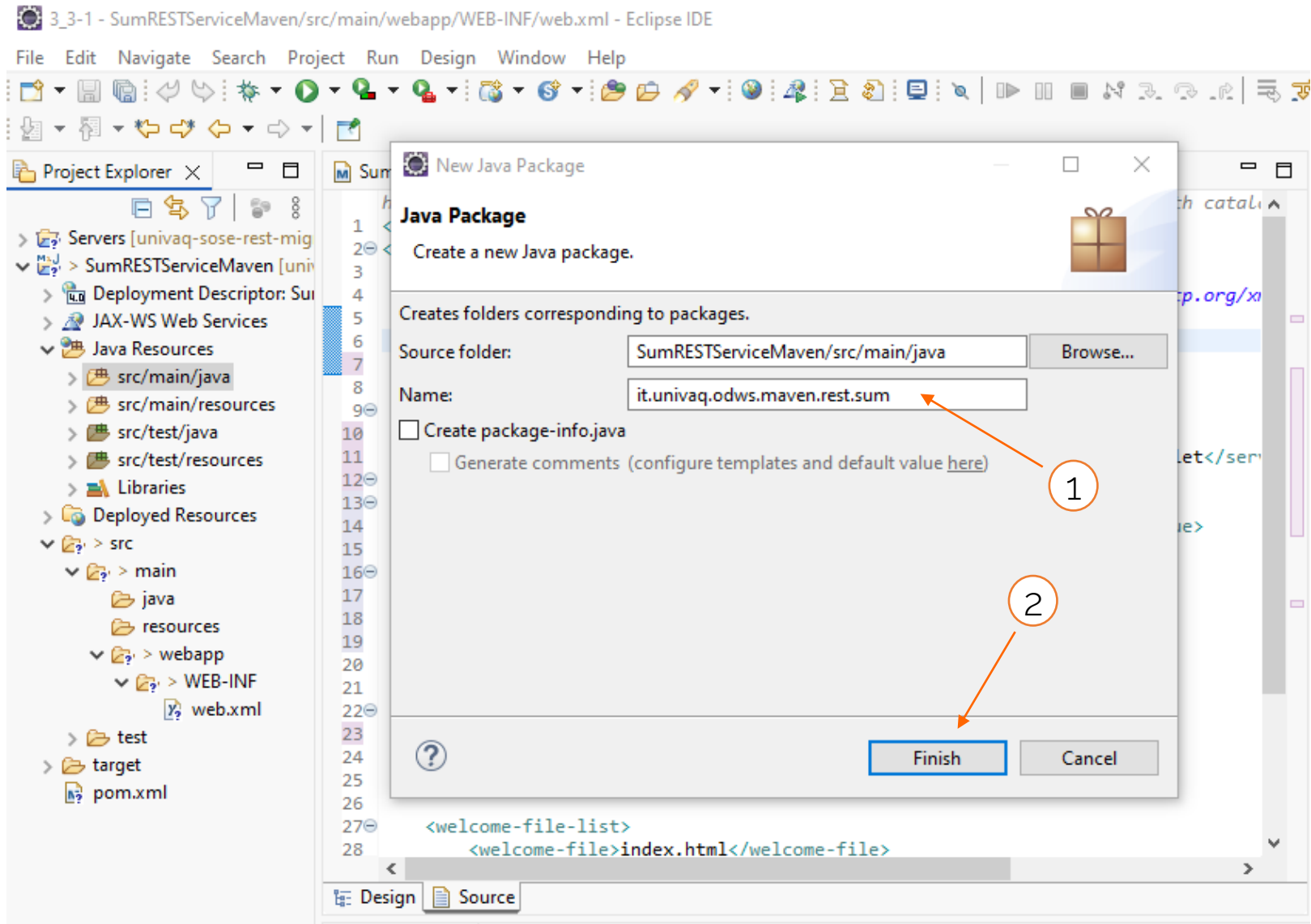
Service Development

Package Creation



Service Development

Package Creation (2)



Service Development

Sum Interface Creation with QueryParam

1

2

```
1 package it.univaq.odws.maven.rest.sum;
2
3 import jakarta.ws.rs.GET;
4 import jakarta.ws.rs.Path;
5 import jakarta.ws.rs.Produces;
6 import jakarta.ws.rs.QueryParam;
7 import jakarta.ws.rs.core.MediaType;
8
9 @Path("/sumrestservice")
10 public interface Sum {
11
12     @GET
13     @Path("/sum")
14     @Produces({MediaType.TEXT_PLAIN})
15     int sum(@QueryParam("a") int a,
16            @QueryParam("b") int b);
17
18 }
19
```

@QueryParam permits to invoke the method with

<http://localhost:8080/SumRESTServiceMaven/rest/sumrestservice/sum?a=7&b=10>

OR

<http://localhost:8080/SumRESTServiceMaven/rest/sumrestservice/sum?b=10&a=7>

Service Development

SumImpl Class Creation

1

2

3

Should you get errors, do not forget to update the Maven project:
pom.xml -> Maven -> Update Project

```
1 package it.univaq.odws.maven.rest.sum;
2
3 public class SumImpl implements Sum {
4
5     @Override
6     public int sum(int a, int b) {
7         return a + b;
8     }
9
10 }
11
```

Service Development

Service Deployment

The screenshot displays an IDE interface for a Java project named `SumRESTServiceMaven`. The **Project Explorer** on the left shows the project structure, including `src/main/java` and `src/main/resources`. The **Outline** on the right shows the `SumImpl` class with the `sum(int, int) : int` method. The main editor shows the `SumImpl.java` file with the following code:

```
1 package it.univaq.odws.maven.rest.sum;
2
3 public class SumImpl implements Sum {
4
5     @Override
6     public int sum(int a, int b) {
7         return a + b;
8     }
9
10 }
11
```

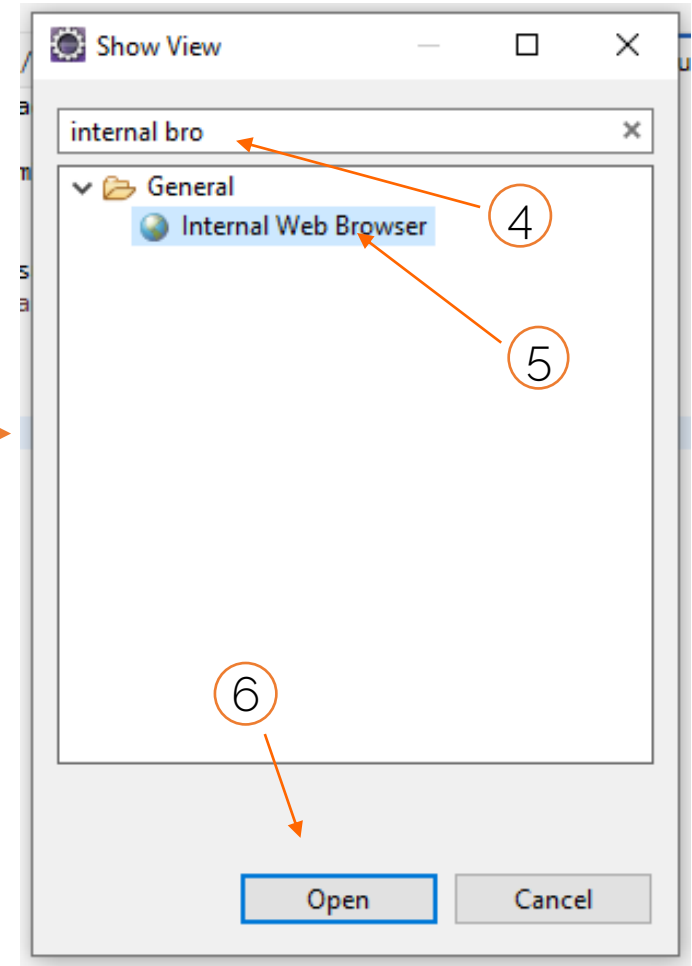
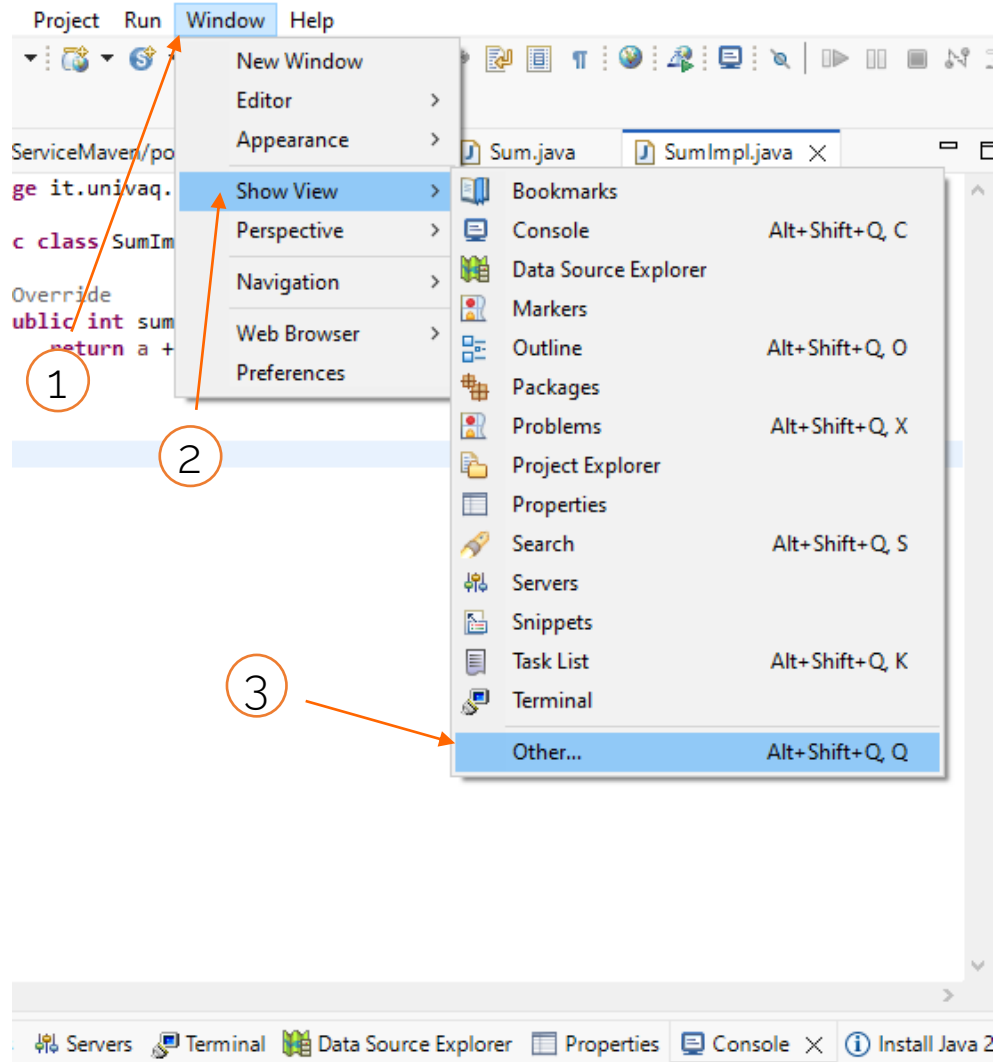
The **Servers** tab at the bottom shows a `Tomcat v11.0 Server at localhost` which is `[Stopped, Synchronized]`. The `SumRESTServiceMaven` project is listed as `[Synchronized]`. A red circle with the number 1 points to the **Start the server** button in the bottom right corner.

The service has been already added to the server in the slide 22

Service Development

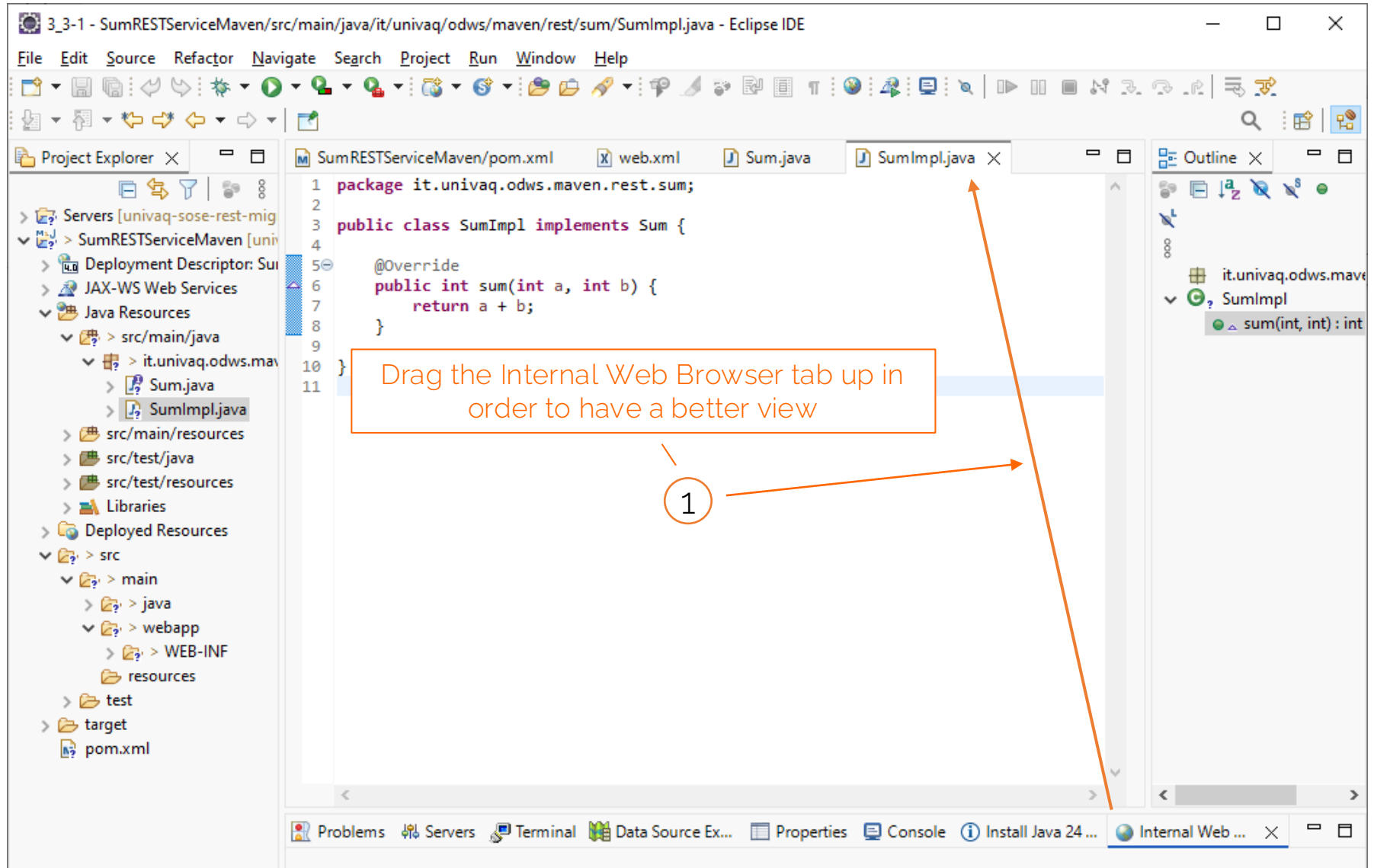
Service Visualization on the Browser

./univaq/odws/maven/rest/sum/SumImpl.java - Eclipse IDE



Service Development

Service Visualization on the Browser (2)



Service Development

Service Visualization on the Browser (3)

The screenshot shows the Eclipse IDE interface. On the left is the Project Explorer showing a project structure with folders like 'src/main/java' and 'src/main/resources'. The main editor area displays a web browser window titled 'Internal Web ...'. The browser's address bar shows the URL 'http://localhost:8080/SumRESTServiceMaven/'. Below the address bar, the page content displays 'Available RESTful services:' followed by a box containing the following information:

- Endpoint address:** <http://localhost:8080/SumRESTServiceMaven/rest>
- WADL:** http://localhost:8080/SumRESTServiceMaven/rest?_wadl

Two orange arrows point to the WADL URL in the browser window. Arrow 1 points to the text 'The service 'home page' will be displayed. Click on the wadl url to check it.' which is enclosed in a dashed orange box. Arrow 2 points to the WADL URL itself. A third orange arrow points from the text box to the 'Endpoint address' line in the browser window.

Service Development

Service Visualization on the Browser (4)

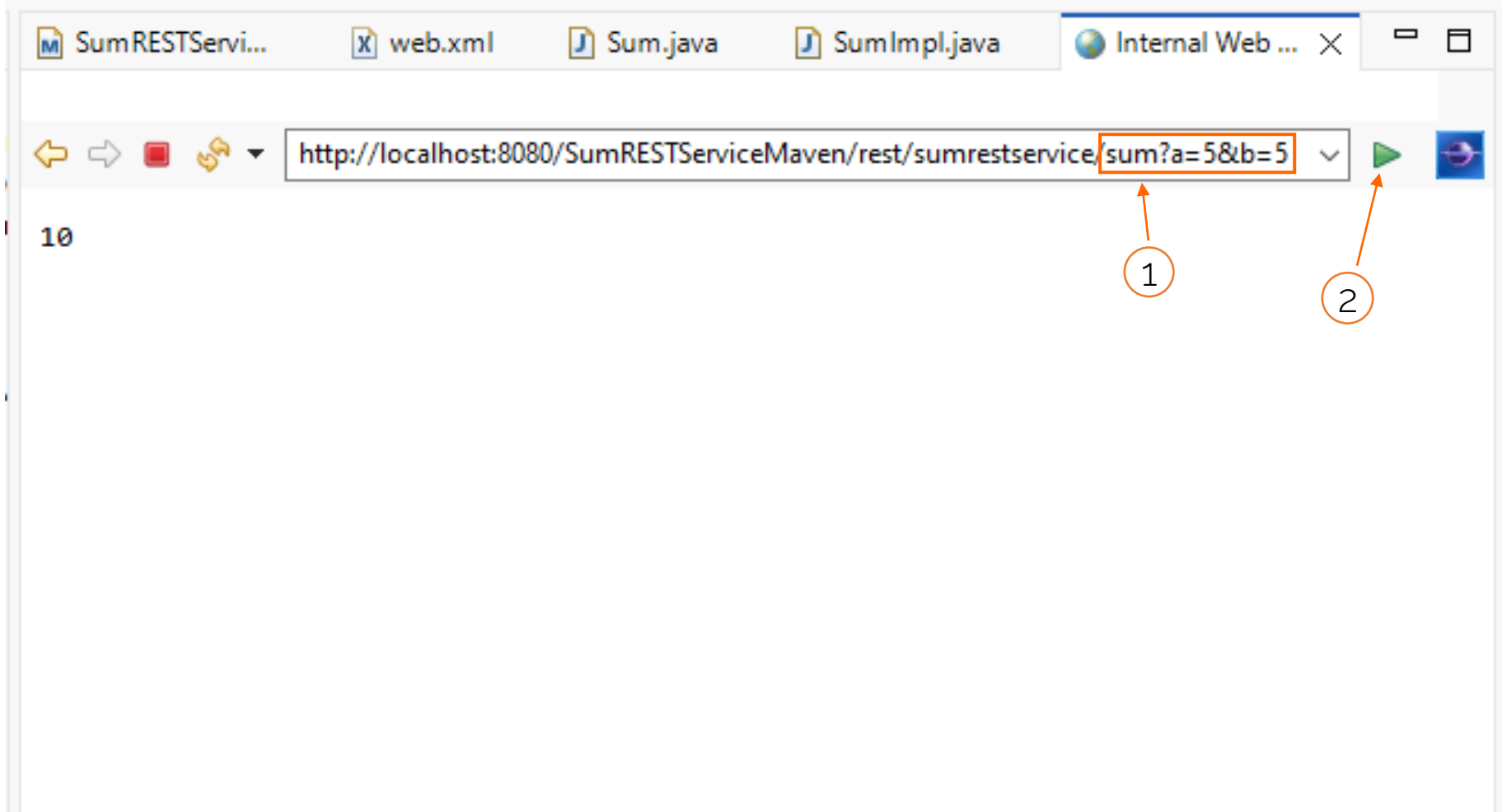
This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8"?>
<application xmlns="http://wadl.dev.java.net/2009/02"
  xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <grammars/>
  <resources base="http://localhost:8080/SumRESTServiceMaven/rest">
    <resource path="/sumrestservice">
      <resource path="/sum">
        <method name="GET">
          <request>
            <param name="a" style="query" type="xs:int"/>
            <param name="b" style="query" type="xs:int"/>
          </request>
          <response>
            <representation mediaType="text/plain">
              <param name="result" style="plain" type="xs:int"/>
            </representation>
          </response>
        </method>
      </resource>
    </resources>
  </application>
```

The service is up and running

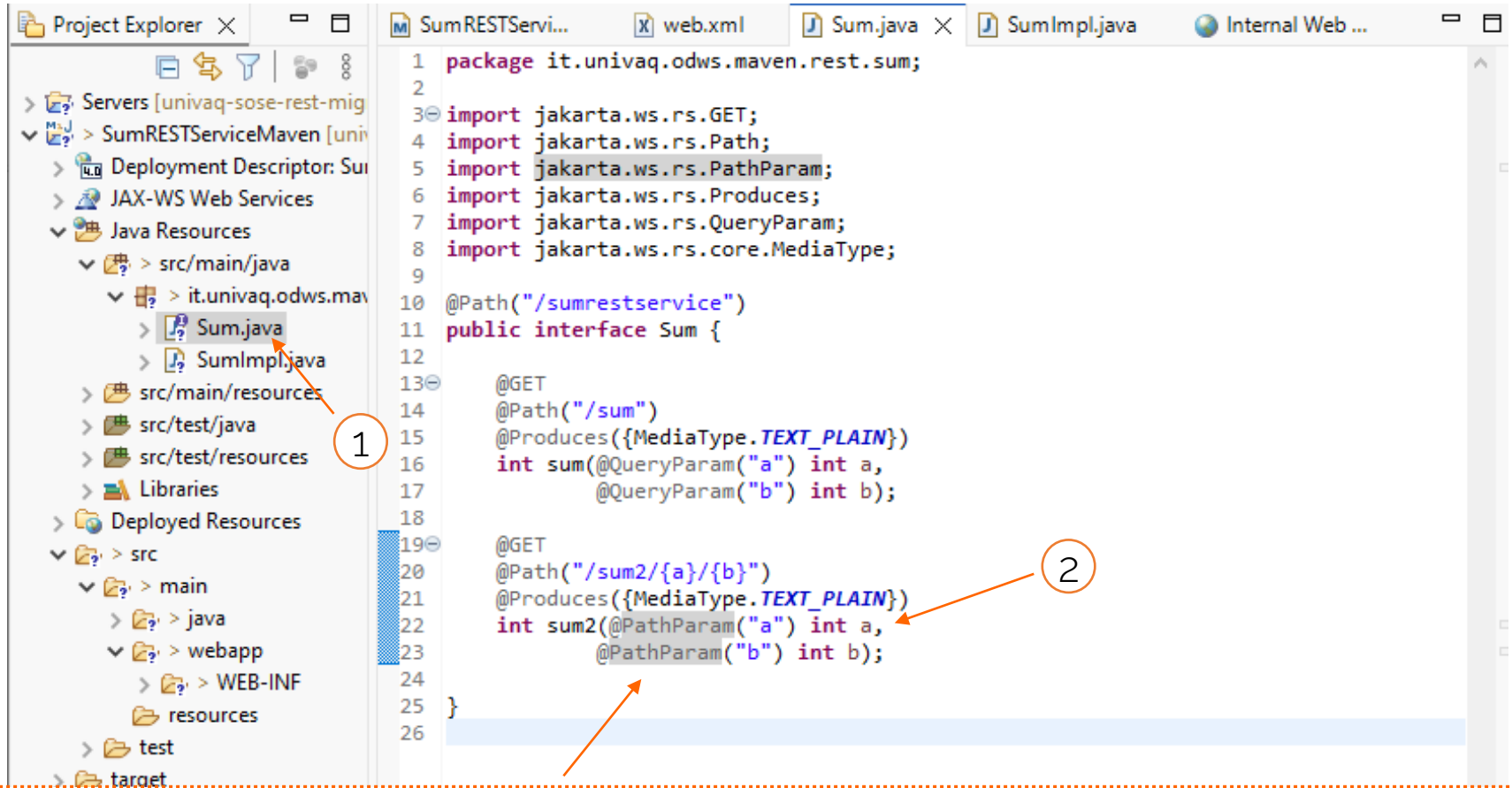
Service Development

Service Test



Service Development

Sum Interface Creation with PathParam



@PathParam permits to invoke the method with
`http://localhost:8080/SumRESTServiceMaven/rest/sumrestservice/sum2/5/5`
much better, isn't it?

Service Development

Sum Interface Creation with PathParam (2)

The screenshot shows the Eclipse IDE interface. The Project Explorer on the left displays the project structure, with the file `SumImpl.java` selected under the path `src/main/java/it.univaq.odws.maven.rest.sum`. The Editor on the right shows the code for `SumImpl.java`, which implements the `Sum` interface. The code is as follows:

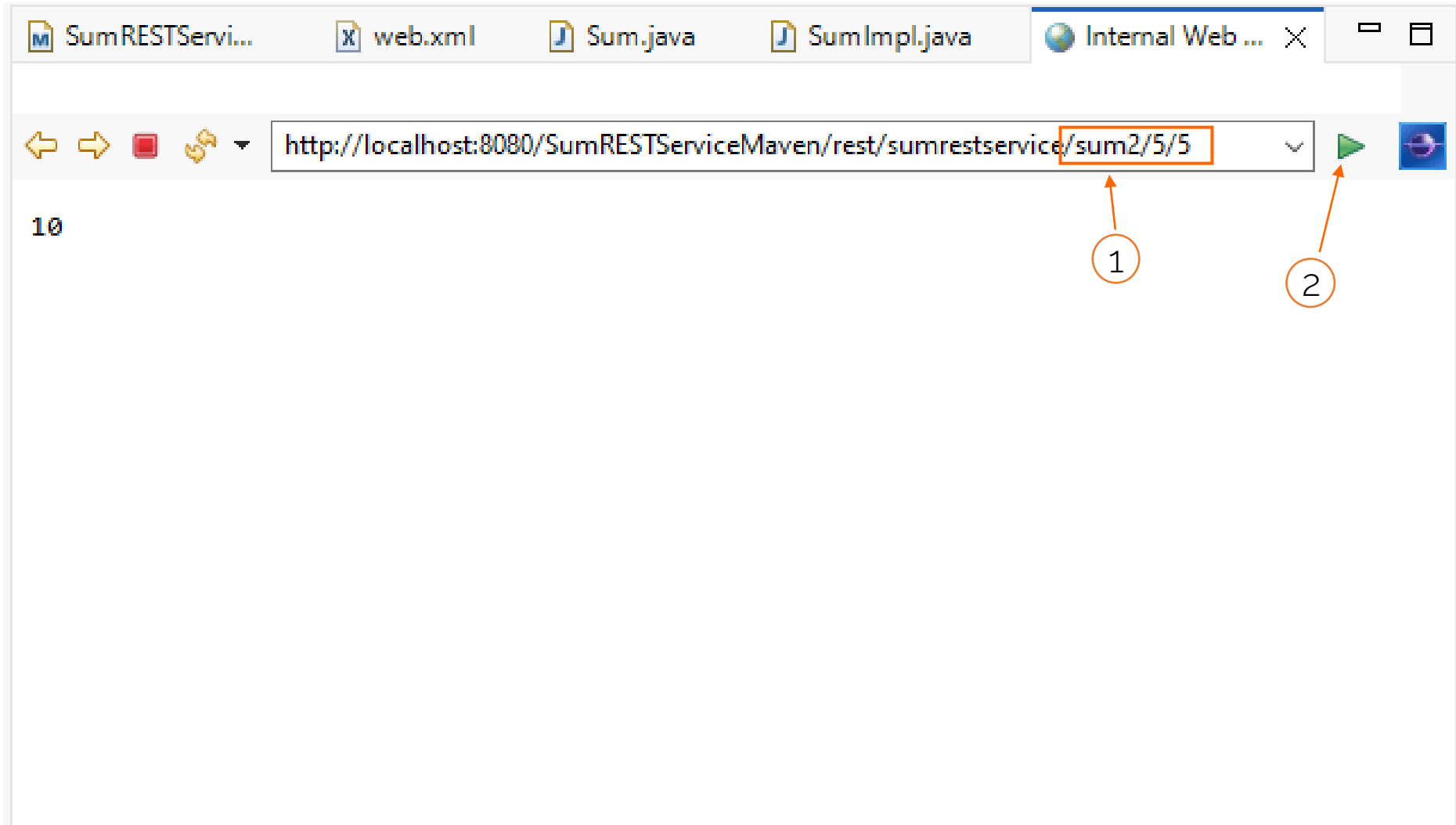
```
1 package it.univaq.odws.maven.rest.sum;
2
3 public class SumImpl implements Sum {
4
5     @Override
6     public int sum(int a, int b) {
7         return a + b;
8     }
9
10    @Override
11    public int sum2(int a, int b) {
12        return a + b;
13    }
14
15 }
16
```

Annotations 1, 2, and 3 are present:

- Annotation 1 points to the `SumImpl.java` file in the Project Explorer.
- Annotation 2 points to the `sum2` method in the code.
- Annotation 3 points to a text box that says "Stop and restart the server".

Service Development

Service Test



Create a Java client for our REST
service

Client Development

New Maven Project

New Maven Project
Configure project

Artifact

Group Id:

Artifact Id:

Version:

Packaging:

Name:

Description:

Parent Project

Group Id:

Artifact Id:

Version:

Create the new project
as in the slides 12-14

Client Development

Pom Configuration (1)

3_3-1 - SumRESTClientMaven/pom.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer

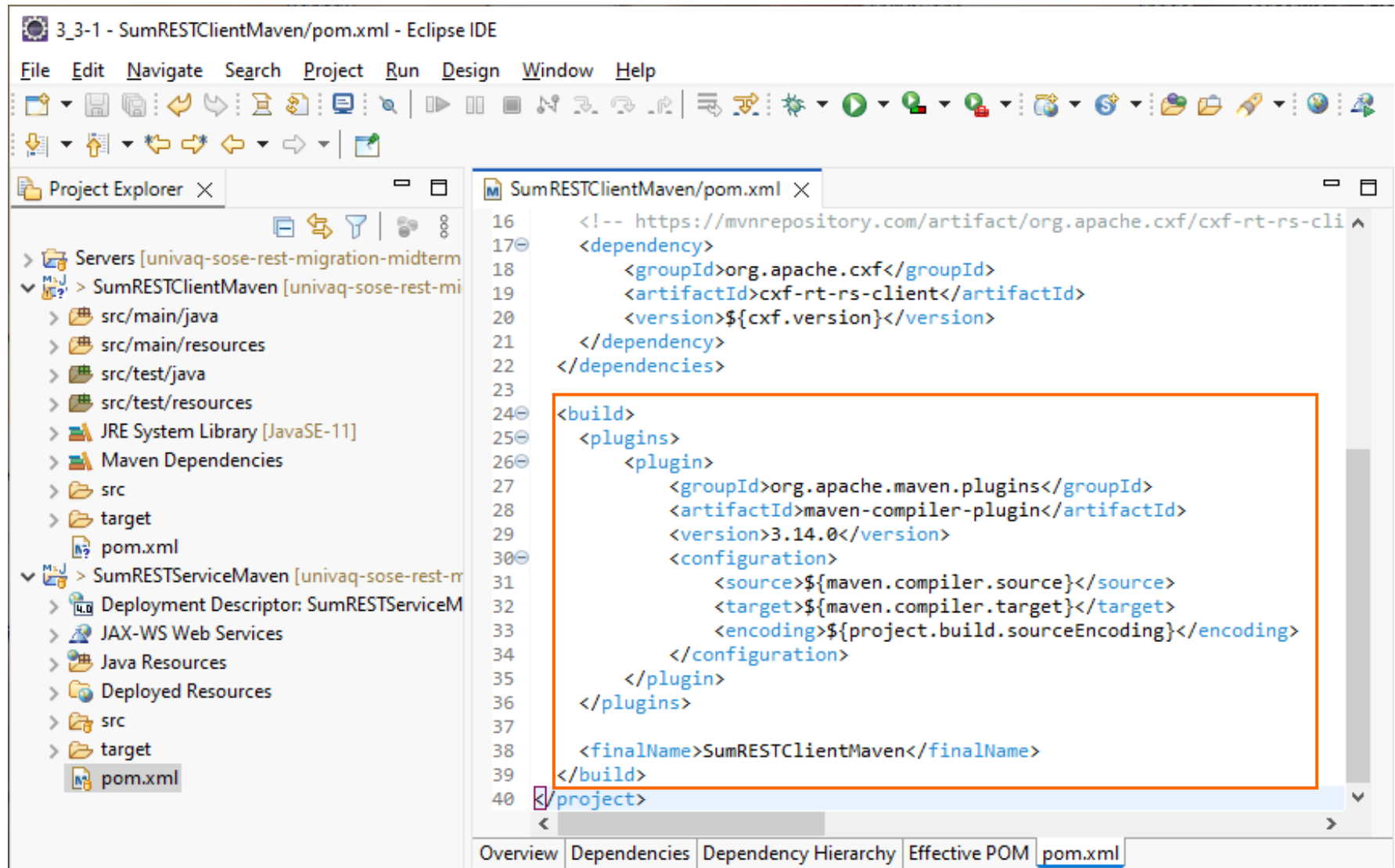
- ▼ GreetingRESTServiceMaven [univaq-sose-rest-m...
- Deployment Descriptor: GreetingRESTServiceM...
- JAX-WS Web Services
- Java Resources
- Deployed Resources
- src
- target
- pom.xml
- Servers [univaq-sose-rest-migration-midterm mai...
- ▼ SumRESTClientMaven [univaq-sose-rest-migrati...
- src/main/java
- src/main/resources
- src/test/java
- src/test/resources
- JRE System Library [JavaSE-1.8]
- Maven Dependencies
- src
- target
- pom.xml
- ▼ SumRESTServiceMaven [univaq-sose-rest-migra...
- Deployment Descriptor: SumRESTServiceMaver...
- JAX-WS Web Services
- Java Resources
- Deployed Resources
- src

```
1 <?xml version="1.0" encoding="UTF-8" ?>
2 <!-- https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation) -->
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4       <groupId>it.univaq.odws.maven.rest.sum.client</groupId>
5       <artifactId>SumRESTClientMaven</artifactId>
6       <version>0.0.1-SNAPSHOT</version>
7
8   <properties>
9     <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
10    <cxf.version>4.1.1</cxf.version>
11    <maven.compiler.source>21.0.6</maven.compiler.source>
12    <maven.compiler.target>21.0.6</maven.compiler.target>
13  </properties>
14
15  <dependencies>
16    <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-rs-client -->
17    <dependency>
18      <groupId>org.apache.cxf</groupId>
19      <artifactId>cxf-rt-rs-client</artifactId>
20      <version>${cxf.version}</version>
21    </dependency>
22  </dependencies>
23
24  <build>
25    <plugins>
26      <plugin>
27        <groupId>org.apache.maven.plugins</groupId>
28        <artifactId>maven-compiler-plugin</artifactId>
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

Client Development

Pom Configuration (2)



3_3-1 - SumRESTClientMaven/pom.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer

- Servers [univaq-sose-rest-migration-midterm]
- SumRESTClientMaven [univaq-sose-rest-mi]
 - src/main/java
 - src/main/resources
 - src/test/java
 - src/test/resources
 - JRE System Library [JavaSE-11]
 - Maven Dependencies
 - src
 - target
 - pom.xml
- SumRESTServiceMaven [univaq-sose-rest-r]
 - Deployment Descriptor: SumRESTServiceM
 - JAX-WS Web Services
 - Java Resources
 - Deployed Resources
 - src
 - target
 - pom.xml

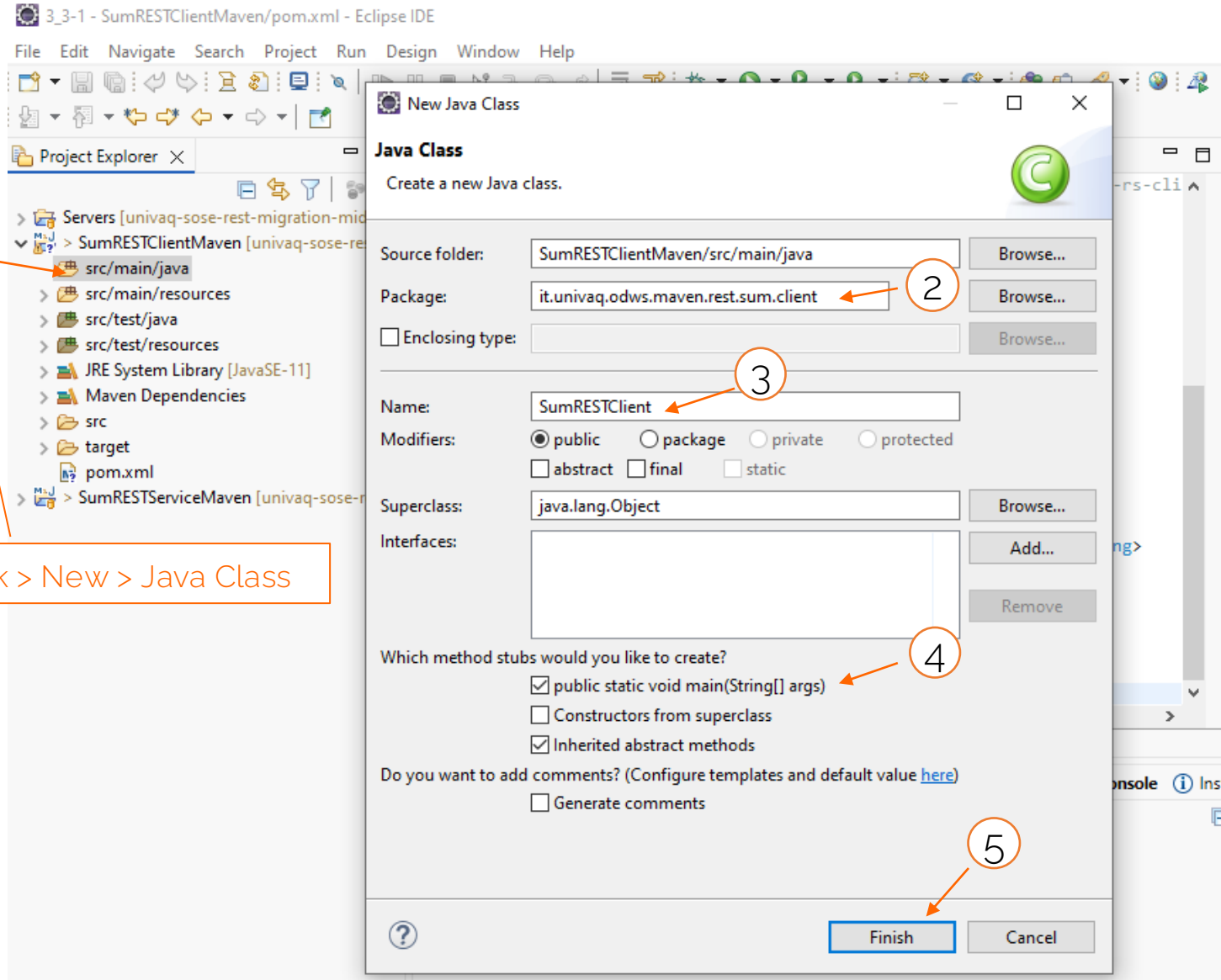
SumRESTClientMaven/pom.xml

```
16 <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-rs-client -->
17 <dependency>
18     <groupId>org.apache.cxf</groupId>
19     <artifactId>cxf-rt-rs-client</artifactId>
20     <version>${cxf.version}</version>
21 </dependency>
22 </dependencies>
23
24 <build>
25     <plugins>
26         <plugin>
27             <groupId>org.apache.maven.plugins</groupId>
28             <artifactId>maven-compiler-plugin</artifactId>
29             <version>3.14.0</version>
30             <configuration>
31                 <source>${maven.compiler.source}</source>
32                 <target>${maven.compiler.target}</target>
33                 <encoding>${project.build.sourceEncoding}</encoding>
34             </configuration>
35         </plugin>
36     </plugins>
37
38     <finalName>SumRESTClientMaven</finalName>
39 </build>
40 </project>
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

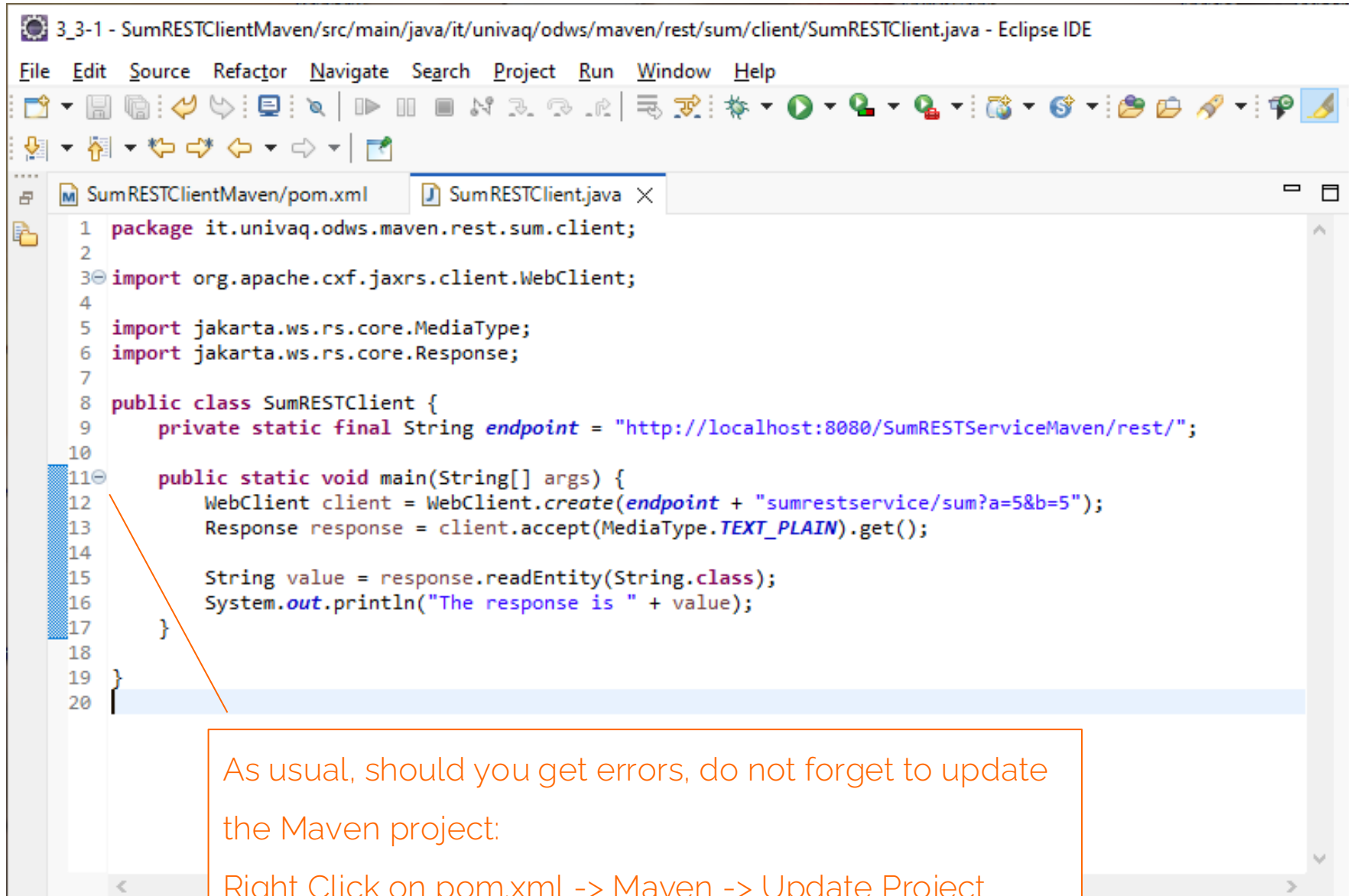
Client Development

New Java Class



Client Development

New Java Class (2)



3_3-1 - SumRESTClientMaven/src/main/java/it/univaq/odws/maven/rest/sum/client/SumRESTClient.java - Eclipse IDE

File Edit Source Refactor Navigate Search Project Run Window Help

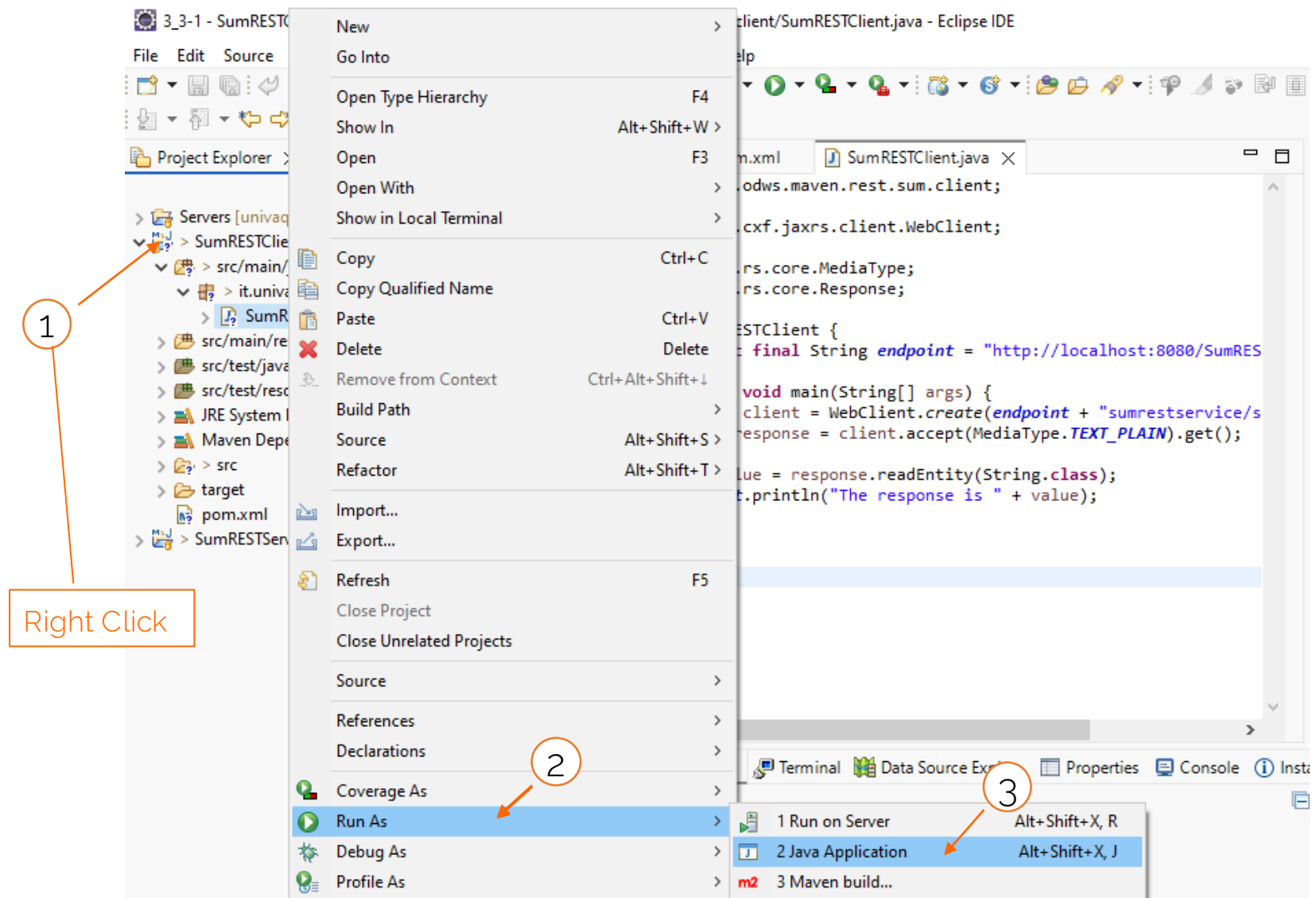
```
1 package it.univaq.odws.maven.rest.sum.client;
2
3 import org.apache.cxf.jaxrs.client.WebClient;
4
5 import jakarta.ws.rs.core.MediaType;
6 import jakarta.ws.rs.core.Response;
7
8 public class SumRESTClient {
9     private static final String endpoint = "http://localhost:8080/SumRESTServiceMaven/rest/";
10
11     public static void main(String[] args) {
12         WebClient client = WebClient.create(endpoint + "sumrestservice/sum?a=5&b=5");
13         Response response = client.accept(MediaType.TEXT_PLAIN).get();
14
15         String value = response.readEntity(String.class);
16         System.out.println("The response is " + value);
17     }
18 }
19
20
```

As usual, should you get errors, do not forget to update the Maven project:

Right Click on pom.xml -> Maven -> Update Project

Client Development

Client Execution



Client Development

Client Execution (2)

3_3-1 - SumRESTClientMaven/src/main/java/it/univaq/odws/maven/rest/sum/client/SumRESTClient.java - Eclipse IDE

File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer

- Servers [univaq-ose-rest-migration-midterm]
- SumRESTClientMaven [univaq-ose-rest-mi]
 - src/main/java
 - it.univaq.odws.maven.rest.sum.client
 - SumRESTClient.java
 - src/main/resources
 - src/test/java
 - src/test/resources
 - JRE System Library [JavaSE-11]
 - Maven Dependencies
 - src
 - target
 - pom.xml
- SumRESTServiceMaven [univaq-ose-rest-n]

```
1 package it.univaq.odws.maven.rest.sum.client;
2
3 import org.apache.cxf.jaxrs.client.WebClient;
4
5 import jakarta.ws.rs.core.MediaType;
6 import jakarta.ws.rs.core.Response;
7
8 public class SumRESTClient {
9     private static final String endpoint = "http://localhost:8080/SumRES
10
11     public static void main(String[] args) {
12         WebClient client = WebClient.create(endpoint + "sumrestservice/s
13         Response response = client.accept(MediaType.TEXT_PLAIN).get();
14
15         String value = response.readEntity(String.class);
16         System.out.println("The response is " + value);
17     }
18 }
19
20
```

1

Problems Servers Terminal Data Source Explo... Properties Console × Insta

<terminated> SumRESTClient [Java Application] D:\Program Files (x64)\Java Versions\jdk-21\bin\javaw.exe (3

The response is 10

Playing with different Media Type in
@Produces annotation

Media Types

Package Creation

The screenshot shows the Eclipse IDE interface with the following components:

- Project Explorer (Left):** Displays the project structure. A callout labeled '2' points to the 'pom.xml' file in the 'target' directory.
- Editor (Right):** Displays the 'pom.xml' file content. A callout labeled '1' points to the top of the editor, and a callout labeled '3' points to the '<artifactId>GreetingRESTServiceMaven</artifactId>' line.
- Callouts:**
 - Callout 1: 'Create the new maven project' (points to the top of the editor).
 - Callout 2: 'Create the Greeting interface and the GreetingImpl class' (points to the 'pom.xml' file in the Project Explorer).
 - Callout 3: Points to the '<artifactId>GreetingRESTServiceMaven</artifactId>' line in the pom.xml.

3_3-1 - GreetingRESTServiceMaven/pom.xml - Eclipse IDE

File Edit Navigate Search Project Run Design Window Help

Project Explorer X

- ▼ GreetingRESTServiceMaven [univaq-rose-rest-mig]
- > Deployment Descriptor: GreetingRESTServiceMaven
- > JAX-WS Web Services
- ▼ Java Resources
- ▼ src/main/java
- ▼ it.univaq.odws.maven.rest.greeting
- > Greeting.java
- > GreetingImpl.java
- > src/main/resources
- > src/test/java
- > src/test/resources
- > Libraries
- > Deployed Resources
- > src
- > target
- pom.xml
- > Servers [univaq-rose-rest-migration-midterm mai]
- > SumRESTClientMaven [univaq-rose-rest-migration]
- > SumRESTServiceMaven [univaq-rose-rest-migration]

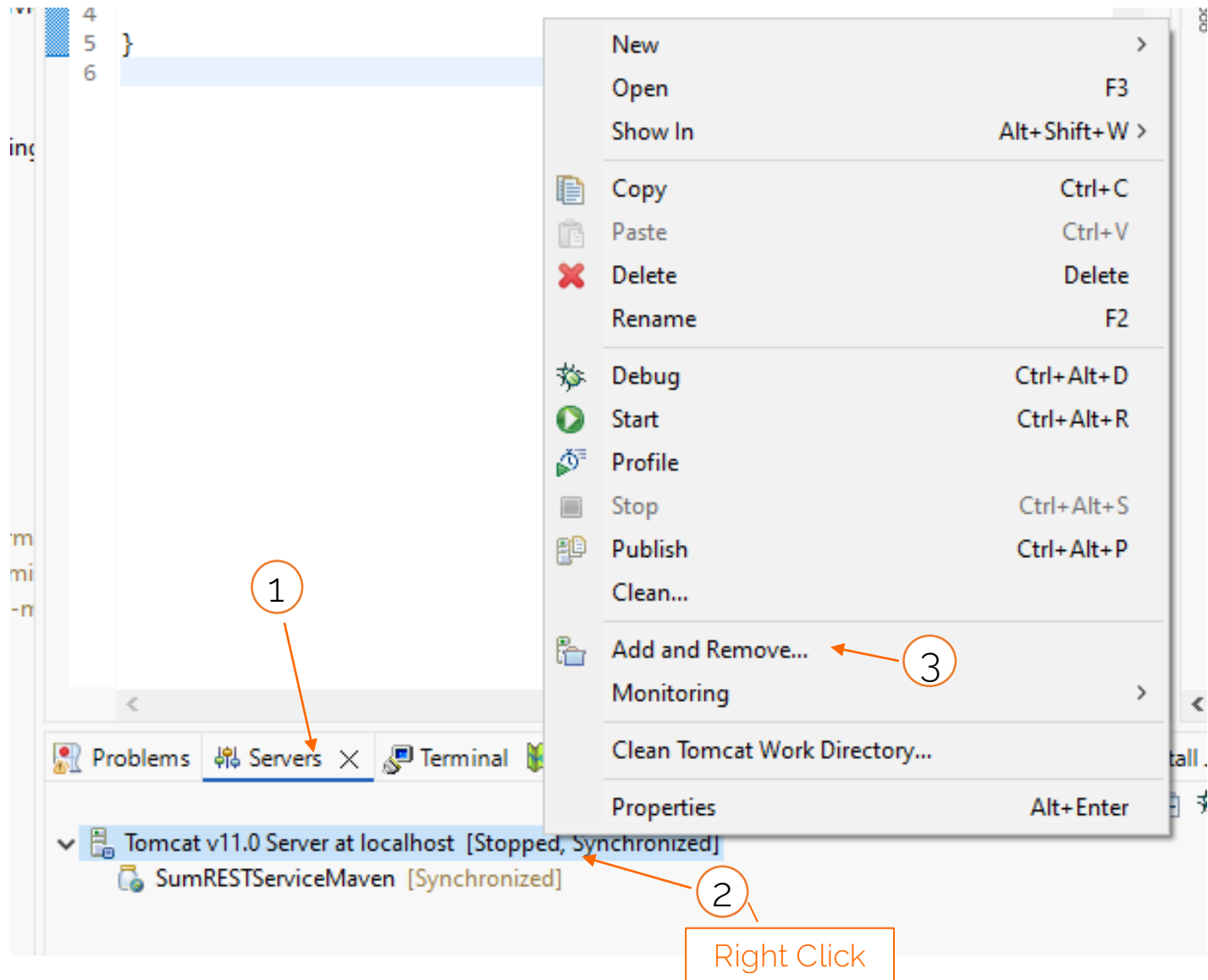
GreetingRESTServiceMaven/pom.xml X

```
2 <modelVersion>4.0.0</modelVersion>
3
4 <groupId>it.univaq.odws.maven.rest.greeting</groupId>
5 <artifactId>GreetingRESTServiceMaven</artifactId>
6 <version>0.0.1-SNAPSHOT</version>
7 <packaging>war</packaging>
8
9 <properties>
10 <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
11 <cxfr.version>4.1.1</cxfr.version>
12 <jakarta.servlet-api>6.1.0</jakarta.servlet-api>
13 <maven.compiler.source>21.0.6</maven.compiler.source>
14 <maven.compiler.target>21.0.6</maven.compiler.target>
15 </properties>
16
17 <dependencies>
18 <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-fronte
19 <dependency>
20 <groupId>org.apache.cxf</groupId>
21 <artifactId>cxf-rt-frontend-jaxrs</artifactId>
22 <version>${cxfr.version}</version>
23 </dependency>
24 <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-tran
25 <dependency>
26 <groupId>org.apache.cxf</groupId>
27 <artifactId>cxf-rt-transports-http</artifactId>
28 <version>${cxfr.version}</version>
29 </dependency>
30 <!-- https://mvnrepository.com/artifact/org.apache.cxf/cxf-rt-rs-
31 <dependency>
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

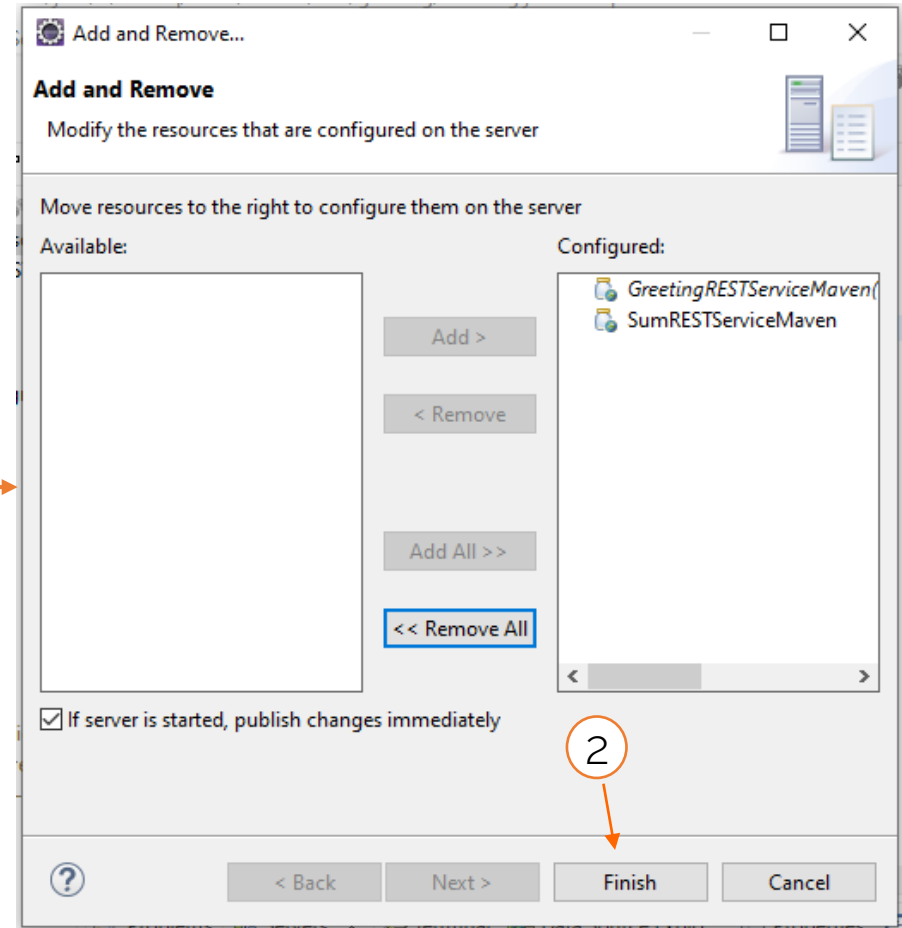
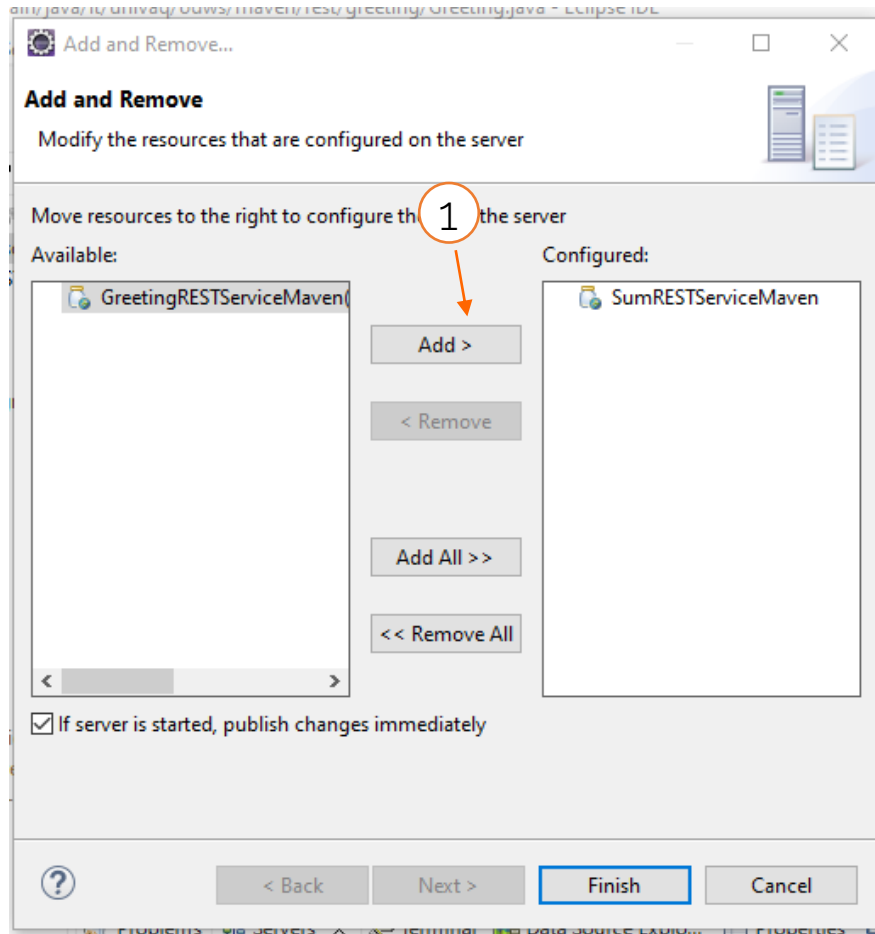
Media Types

Service Addition to the Server



Media Types

Service Addition to the Server (2)



Media Types

Greeting Interface

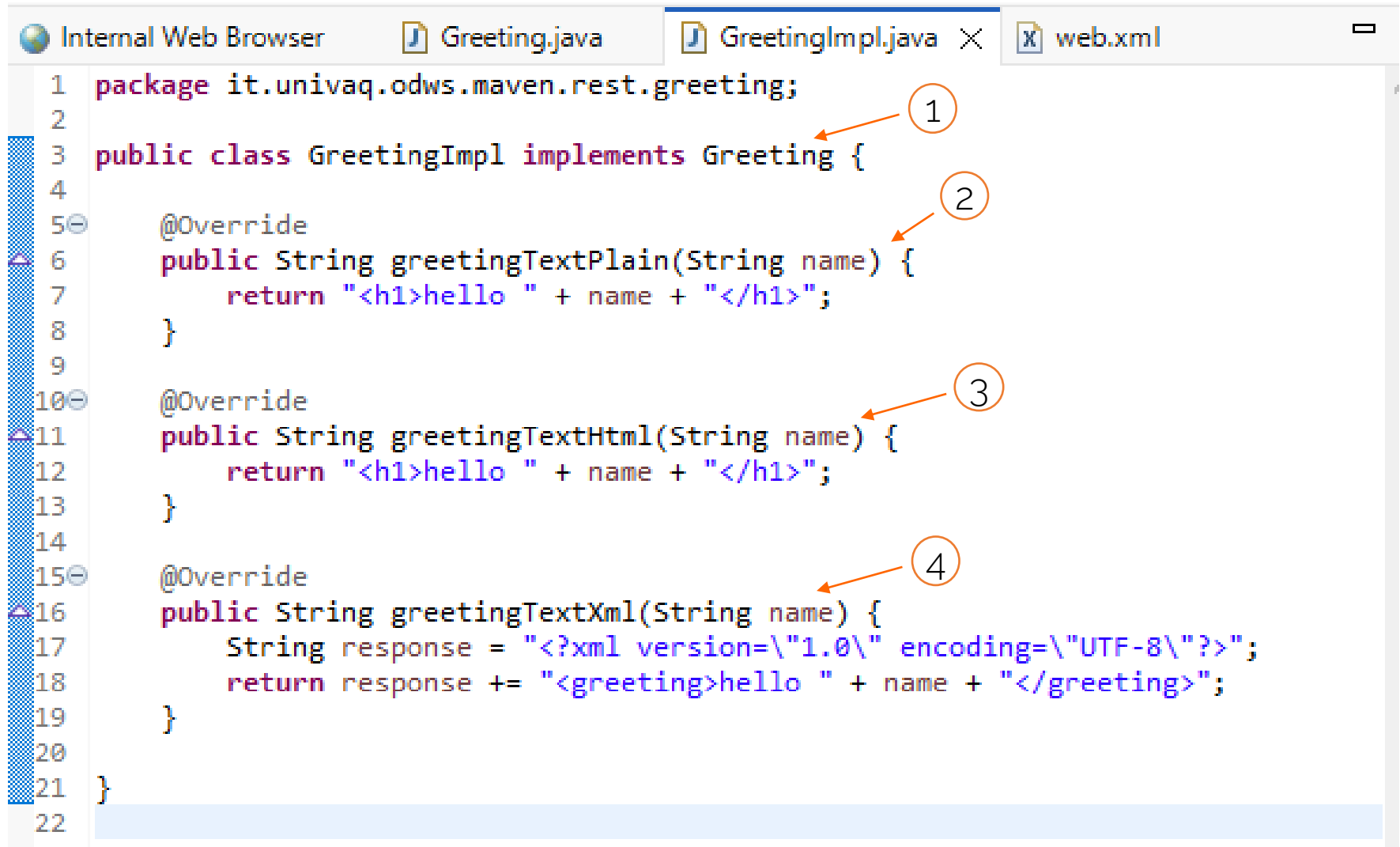
```
Greeting.java X GreetingImpl.java Sum.java
1 package it.univaq.odws.maven.rest.greeting;
2
3 import jakarta.ws.rs.GET;
4 import jakarta.ws.rs.Path;
5 import jakarta.ws.rs.PathParam;
6 import jakarta.ws.rs.Produces;
7 import jakarta.ws.rs.core.MediaType;
8
9 @Path("/greetingrestservice")
10 public interface Greeting {
11
12     @GET
13     @Path("/greeting_text_plain/{name}")
14     @Produces({ MediaType.TEXT_PLAIN })
15     String greetingTextPlain(@PathParam("name") String name);
16
17     @GET
18     @Path("/greeting_text_html/{name}")
19     @Produces({ MediaType.TEXT_HTML })
20     String greetingTextHtml(@PathParam("name") String name);
21
22     @GET
23     @Path("/greeting_text_xml/{name}")
24     @Produces({ MediaType.TEXT_XML })
25     String greetingTextXml(@PathParam("name") String name);
26
27 }
28
```

The `@Produces` annotation is used to specify the media types or representations a resource can produce and send back to the client

e.g., `TEXT_PLAIN` represents the String constant `text/plain` media type

Media Types

Greeting Class Implementation



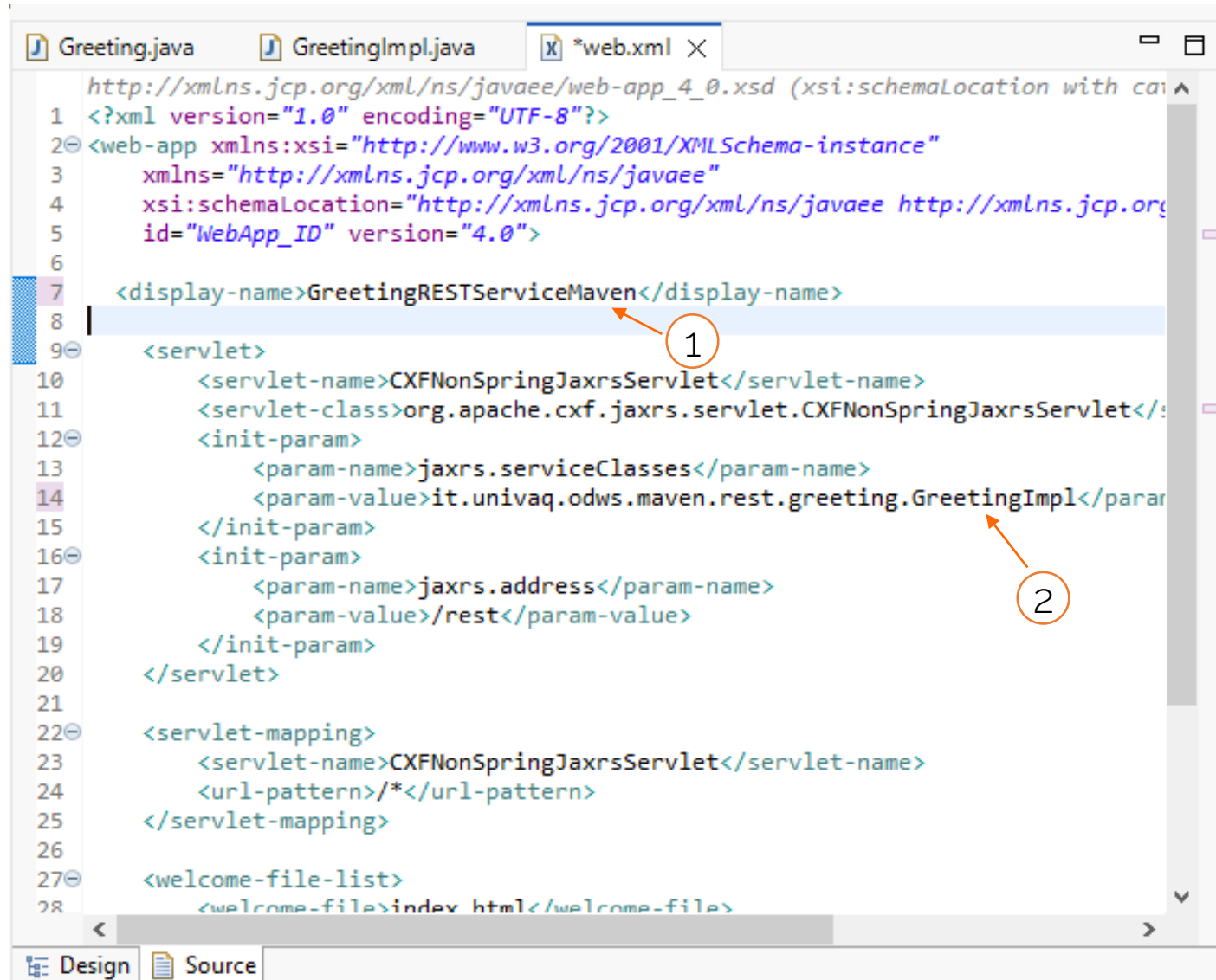
```
1 package it.univaq.odws.maven.rest.greeting;
2
3 public class GreetingImpl implements Greeting {
4
5     @Override
6     public String greetingTextPlain(String name) {
7         return "<h1>hello " + name + "</h1>";
8     }
9
10    @Override
11    public String greetingTextHtml(String name) {
12        return "<h1>hello " + name + "</h1>";
13    }
14
15    @Override
16    public String greetingTextXml(String name) {
17        String response = "<?xml version=\"1.0\" encoding=\"UTF-8\"?>";
18        return response += "<greeting>hello " + name + "</greeting>";
19    }
20
21 }
22
```

Annotations in the image:

- 1: Points to the `implements Greeting` part of the class declaration.
- 2: Points to the `greetingTextPlain` method signature.
- 3: Points to the `greetingTextHtml` method signature.
- 4: Points to the `greetingTextXml` method signature.

Media Types

Servlet Configuration



The screenshot shows an IDE window with three tabs: Greeting.java, GreetingImpl.java, and *web.xml. The web.xml file is open and displays the following XML code:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <web-app xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3       xmlns="http://xmlns.jcp.org/xml/ns/javaee"
4       xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org
5       id="WebApp_ID" version="4.0">
6
7   <display-name>GreetingRESTServiceMaven</display-name>
8
9   <servlet>
10     <servlet-name>CXFNonSpringJaxrsServlet</servlet-name>
11     <servlet-class>org.apache.cxf.jaxrs.servlet.CXFNonSpringJaxrsServlet</servlet-class>
12     <init-param>
13       <param-name>jaxrs.serviceClasses</param-name>
14       <param-value>it.univaq.odws.maven.rest.greeting.GreetingImpl</param-value>
15     </init-param>
16     <init-param>
17       <param-name>jaxrs.address</param-name>
18       <param-value>/rest</param-value>
19     </init-param>
20   </servlet>
21
22   <servlet-mapping>
23     <servlet-name>CXFNonSpringJaxrsServlet</servlet-name>
24     <url-pattern>/*</url-pattern>
25   </servlet-mapping>
26
27   <welcome-file-list>
28     <welcome-file>index.html</welcome-file>
29   </welcome-file-list>
```

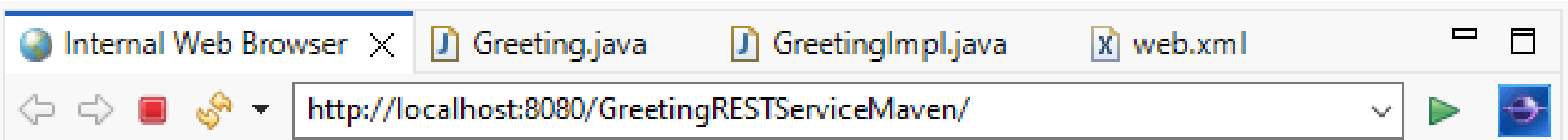
Two annotations are present in the code:

- Annotation 1 (a circle with the number 1) points to the `<servlet>` tag on line 9.
- Annotation 2 (a circle with the number 2) points to the `<param-value>it.univaq.odws.maven.rest.greeting.GreetingImpl</param-value>` on line 14.

The IDE interface at the bottom shows the "Design" and "Source" tabs, with "Source" currently selected.

Media Types

Service Testing



Available RESTful services:

Endpoint address: <http://localhost:8080/GreetingRESTServiceMaven/rest>
WADL : http://localhost:8080/GreetingRESTServiceMaven/rest?_wadl

1

Open the Internal Web
Browser as in the slides 36-38

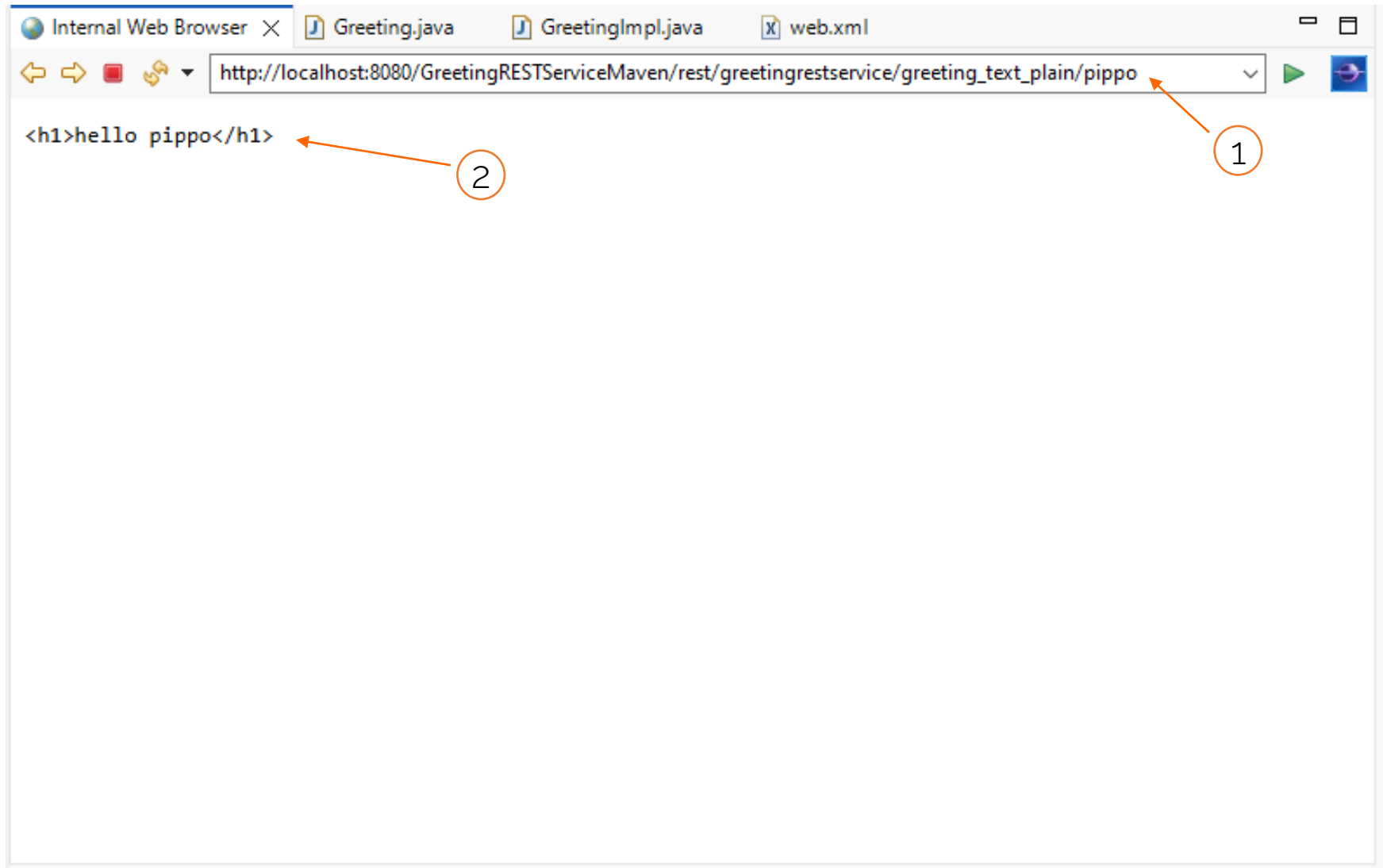
Media Types

Service Testing (2)

```
<grammars/>
<resources base="http://localhost:8080/GreetingRESTServiceMaven/rest">
  <resource path="/greetingrestservice">
    <resource path="/greeting_text_html/{name}"> ← 1
      <param name="name" style="template" type="xs:string"/>
      <method name="GET">
        <request/>
        <response>
          <representation mediaType="text/html"> ← 2
            <param name="result" style="plain" type="xs:string"/>
          </representation>
        </response>
      </method>
    </resource>
    <resource path="/greeting_text_plain/{name}"> ← 3
      <param name="name" style="template" type="xs:string"/>
      <method name="GET">
        <request/>
        <response>
          <representation mediaType="text/plain"> ← 4
            <param name="result" style="plain" type="xs:string"/>
          </representation>
        </response>
      </method>
    </resource>
    <resource path="/greeting_text_xml/{name}"> ← 5
      <param name="name" style="template" type="xs:string"/>
      <method name="GET">
        <request/>
        <response>
```

Media Types

Service Testing with Text Plain MediaType



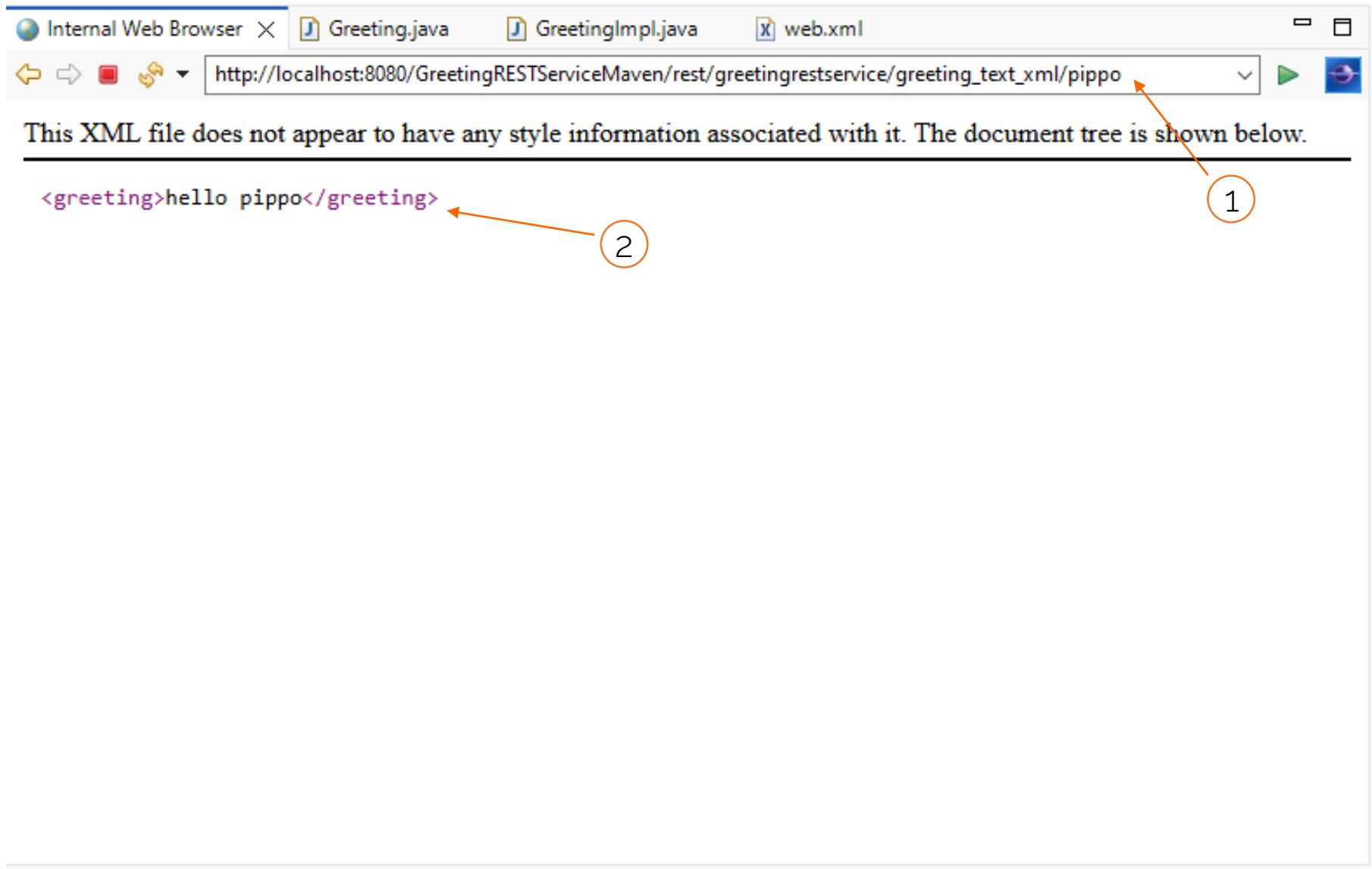
Media Types

Service Testing with HTML MediaType



Media Types

Service Testing with XML MediaType



References

- <https://javaee.github.io/tutorial/jaxrs002.html>
- <https://jakarta.ee/specifications/restful-ws/3.0/apidocs/jakarta/ws/rs/core/mediatype>
- <https://docs.oracle.com/cd/E19798-01/821-1841/gipzh/index.html>
- <http://cxf.apache.org/docs/jax-rs-client-api.html>